# Medical Image Fusion: Fusion Paradigms, Clinical Applications, and Open Resources

Junxin Chen[a], Zhiheng Ye[a], Mingzhi Wang[a], Jiawei Zhang[a], Jun Mou[b,*], Xu Xu[c], Yongfei Wu[d] and Wei William Wang[e]

[a]School of Software, Dalian University of Technology, Dalian, 116621, China
[b]School of Information Science and Engineering, Dalian Polytechnic University, Dalian, 116034, China
[c]School of Computer Science and Engineering, Northeastern University, Shenyang, 110004, China
[d]College of Artificial Intelligence, University of Taiyuan Technology, Taiyuan, 030024, China
[e]Engineering Research Centre of Applied Technology on Machine Translation and Artificial Intelligence, Macao Polytechnic University, Macao SAR, 999078, China

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ABSTRACT

Multi-modality medical image fusion (MMIF) integrates complementary information from different medical imaging modalities or sequences at the data, feature, or decision level to produce fused images or image-based clinical predictions. This review organizes MMIF through a three-axis taxonomy of input composition, fusion stage, and output purpose, and connects this taxonomy with open resources, evaluation, and clinical applications. The review centers on in vivo medical imaging, covering fused-image generation and task-oriented fusion for segmentation, detection, and classification; general fusion of images with clinical records, reports, or omics data lies outside its detailed comparison. Within this framework, we trace the evolution from traditional multi-scale transformation methods to modern deep learning approaches and analyze how different paradigms exploit complementary information while managing redundancy, inconsistency, and uncertainty across modalities. We further summarize an open-resource workflow spanning data preprocessing, model development, evaluation, and deployment to improve reproducibility and narrow the gap between algorithmic innovation and engineering implementation. Key challenges include modality heterogeneity, cross-modality misalignment, uncertainty propagation, and the absence of unified evaluation standards; promising directions include robust representation alignment, uncertainty-aware fusion, and privacy-preserving multi-institutional collaboration. By linking fusion architectures, algorithms, and image-based clinical tasks, this review provides a structured reference for the design and translation of practical MMIF systems. A curated collection of relevant resources and corresponding links is publicly available at: https://github.com/CHamYE/MEDFusion/.

## 1. Introduction

In this review, multi-modality medical image fusion (MMIF) refers to the integration of complementary information from different medical imaging modalities or sequences at the data, feature, or decision level. Its outputs may be fused images or task-specific predictions derived from the imaging sources. Beyond aggregating anatomical, functional, and temporal information, MMIF must preserve complementary evidence, suppress redundancy, resolve cross-source inconsistencies, and account for uncertainty at the data, feature, and decision levels. These characteristics connect MMIF to the broader problems of multi-sensor, multi-source, and multi-process information fusion while making it particularly important for reliable clinical analysis and decision support.

*Corresponding author
thumbnails/cas-email.jpeg; Junxinchen@ieee.org (J. Chen); 175299084@mail.dlut.edu.cn (Z. Ye); wmz@mail.dlut.edu.cn (M. Wang); 2394719167@mail.dlut.edu.cn (J. Zhang); moujun@dlpu.edu.cn (J. Mou); xuxu@ieee.org (X. Xu); wuyongfei@tyut.edu.cn (Y. Wu); weiwang@mpu.edu.mo (W.W. Wang)
ORCID(s): 0000-0003-4745-8361 (J. Chen); 0000-0002-3453-6474 (M. Wang); 0000-0002-7774-2833 (J. Mou); 0000-0002-3934-9096 (X. Xu); 0000-0002-5010-2561 (Y. Wu); 0000-0002-1717-5785 (W.W. Wang)

### 1.1. Objectives

Medical images acquired by different sensors encode distinct but complementary evidence because each modality is governed by different physical principles. CT provides high-resolution anatomical structures, MRI offers strong soft-tissue contrast, and PET/SPECT reveal functional or metabolic activity, but no single modality can provide all the information required for complex clinical decision-making [1, 2, 3]. Their differences in intensity distribution, spatial resolution, noise characteristics, and acquisition geometry make MMIF more than a simple image-combination problem: it requires cross-modality alignment, representation learning, and principled management of complementary, redundant, and potentially conflicting information [4, 5]. These requirements arise in image-based studies of tumor segmentation, lesion detection, and radiotherapy contouring, with different endpoints and validation settings [6, 7, 8].

MMIF addresses these requirements by integrating observations at the data, feature, or decision level according to the intended clinical task [9, 10]. Representative studies evaluate fusion for brain tumor segmentation and tumor contouring [6, 8]. Such task results should be distinguished from evidence of benefit in a deployed clinical workflow. The

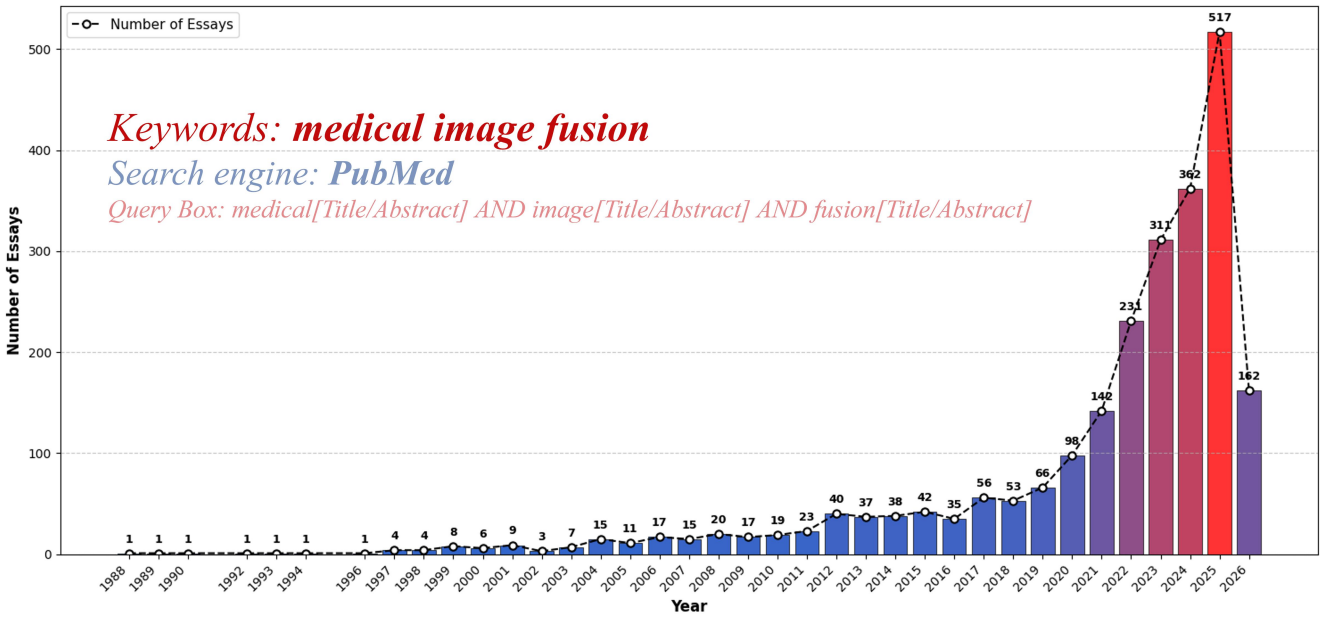



**Figure 1**: Annual PubMed records on medical image fusion, 1988–2026 (retrieved 17 March 2026; the 2026 bar is partial).

growing importance of MMIF is also reflected in the increasing number of PubMed publications containing the keywords "medical[Title/Abstract] AND image[Title/Abstract] AND fusion[Title/Abstract]", as shown in Fig. 1.

Despite this progress, alignment, missing inputs, and modality reliability remain method-specific constraints [11, 12, 3]. This review therefore relates input composition, fusion stage, and output purpose through an explicit three-axis taxonomy, alongside an overview of open resources and clinical applications. **Scope and boundaries.** The detailed comparison centers on in vivo medical imaging, including radiological and other clinical imaging modalities and multi-sequence MRI. Segmentation, detection, and classification studies are included when cross-modality image integration is a central methodological component. Decision-level fusion is included when it combines outputs associated with distinct imaging modalities or sequences; single-modality prediction and generic model ensembles are outside this scope. Missing-modality learning, modality synthesis, and registration are discussed as supporting components of an MMIF workflow, with synthesis distinguished from fusion itself. Text guidance is included only when it assists the fusion of multiple medical images. General image–text diagnosis and fusion with electronic records, clinical tables, or genomic data are outside the detailed comparison. Cross-scale radiology–pathology fusion is an adjacent image-based extension outside the in vivo focus of the detailed comparison.

## 1.2. Differences and contributions

To clarify the positioning of this review, we first summarize its main differences from existing representative surveys. Existing reviews have established substantial foundations for MMIF, including modality characteristics, fusion methods, preprocessing, public datasets, evaluation, and clinical applications. Table 1 compares representative general reviews and task-focused reviews using explicit topic-coverage criteria. The comparison covers a representative selection of surveys and uses their cited full texts or identified public author versions.

**Comparison criteria.** A checkmark (✓) indicates that the specified content is explicitly documented in the text, a table, or a described workflow. A cross (×) indicates that the criterion is not met; passing mentions may still occur. For *modality features*, the review explains imaging properties or complementarity; for *public datasets*, it identifies public image collections and describes their use or access; for *preprocessing*, it describes an input-preparation or registration operation and its purpose or procedure. For *metrics*, it explains, interprets, or compares image-quality or task-performance measures; for *open tools/code*, it identifies reusable software or method implementations with their purpose or availability; and for *challenges or trends*, it discusses limitations or research directions. A software name used only to report an experimental platform, or an item appearing only in the references, is insufficient. T denotes traditional methods, including shallow learning and rule-based neural models; D denotes deep learning; T/D denotes both. Task scope is stated descriptively. These indicators record documented topic coverage. Depth, completeness, methodological quality, and novelty require qualitative comparison.

**Qualitative comparison with prior surveys.** The reviews in Table 1 provide substantial foundations. James and Dasarathy organize their 2014 survey around organ applications and registration [13]; their 2015 account emphasizes feature and data fusion. Our three-axis mapping separates input composition, integration stage, and output

purpose across these settings. Rajalingam and Priya focus on combined transforms; our comparison extends to learned and generative methods under explicit data and cost assumptions. Yadav and Yadav examine hybrid methods, while Venkatesan et al. review a range of fusion approaches [14, 15]; we complement these accounts with implementation-specific comparisons of pairing, missing inputs, supervision, and robustness.

Huang et al. connect public image sources with quantitative evaluation [16]; our evaluation discussion additionally distinguishes image scores from task and reader evidence. Basu et al. analyze decomposition, fusion rules, modality combinations, and metrics [17]; we connect method assumptions to output-specific evaluation and reproduction records. Zhou et al. integrate segmentation strategies, datasets, preprocessing, metrics, and code [18]; our mapping relates segmentation to fused-image generation and other prediction tasks. Li et al. provide a detailed classification-oriented account of fusion schemes, incomplete inputs, evaluation, and resources [19]; our contribution is the explicit comparison of output objectives and their supervision and evaluation requirements across the defined imaging scope.

Azam et al. cover modalities, databases, fusion techniques, and quality measures [1]; we build on this breadth with operational taxonomy rules and structured method-condition comparisons. Khan et al. organize fusion techniques, image sources, quality measures, and implementation links [20]; our reproducibility protocol specifies the records needed at each stage of a selected pipeline. Zubair et al. cover traditional and learned approaches, clinical applications, and trade-offs in robustness, efficiency, and interpretability [21]. This review relates these concerns through a common comparison of method conditions, study-level validation evidence, and explicit evaluation and reproduction requirements. Content comparisons for Zubair et al. use the identified public author version, as recorded in the bibliography.

**Contributions.** Building on established accounts of fusion paradigms, clinical applications, and open resources, this review connects four analytical perspectives:

1) **Operational taxonomy.** An operational taxonomy separates input composition, fusion stage, and output purpose, with explicit assignment rules and representative configurations within the stated imaging scope.

2) **Method conditions.** We compare registration, paired data, missing inputs, supervision, cost, and robustness across representative implementations. These conditions also inform task-specific method selection.

3) **Interpretation of validation evidence.** We distinguish study design from metric performance, synthesize validation evidence and reported quantitative results, and explain disagreements among visual, task, and reader assessments.

4) **Reproduction requirements.** The reproducibility protocol specifies data, preprocessing, implementation, execution, and evaluation records for reconstructing a selected workflow. These records link configuration choices to the corresponding outputs and evaluation results.

### 1.3. Survey methodology

This review is a structured survey rather than a registered systematic review. The search, screening, and counting rules are reported below so that the corpus and the statistics in Fig. 9 can be reconstructed from the released study list.

**Search protocol.** PubMed was the primary database because it covers the core clinical and biomedical literature in this field. We supplemented it with backward and forward citation chasing from the key studies identified during screening. The PubMed query behind Fig. 1 was `medical[Title/Abstract] AND image[Title/Abstract] AND fusion[Title/Abstract]`, executed on 17 March 2026 and restricted to publication dates from 1 January 1988 to that date. The topical search combined three concept blocks: (i) medical, clinical, radiological, or biomedical imaging; (ii) two or more imaging sources, including modalities, sequences, or dynamic phases; and (iii) fusion, integration, or combination. The method corpus covers publications from 1993 to March 2026.

**Screening and eligibility.** Records were screened by reading the title and abstract against the scope terms defined in Section 1.1, and the retained records were assessed on the full text or, when the full text was not accessible, on the publisher record together with its tables and figures. A study was included when it (i) reported a primary method rather than a review, editorial, or resource description; (ii) integrated at least two medical imaging sources at the data, feature, or decision level; (iii) reported at least one quantitative fusion metric, task metric, or structured reader assessment; and (iv) fell within the imaging scope defined in Section 1.1. Studies of single-modality enhancement, image–text or image–omics fusion without multi-image integration, and non-medical infrared–visible fusion were excluded from the corpus. Cross-domain infrared–visible results are cited only as external evidence and are identified as such in the text.

**Corpus.** The corpus contains 192 distinct studies, catalogued in Tables 3–5 and released with per-study extraction in the companion repository (`studies_metric_coding.csv`). The three tables contain 211 entries because 17 studies are listed in more than one configuration and 16 of these span more than one fusion level. Seventy-four studies appeared in 2023–2026, 59 in 2020–2022, 39 in 2015–2019, and 20 before 2015; the earliest is from 1993. Reviews, dataset descriptions, software documentation, and regulatory guidance are cited for context but are not counted in the 192 studies.

**Metric statistics.** Fig. 9 counts the 13 most frequently reported metric families over the corpus. Each study contributes at most one count per family, regardless of the number of datasets, configurations, or variants it reports. Variants are merged at the family level: MI includes NMI and

LMI; SSIM includes MS-SSIM and MSSIM; SD includes STD; EN includes entropy; DSC includes Dice; and Acc includes accuracy and pixel-wise accuracy. Visual assessment, runtime, registration error, and other task-specific measures were recorded in the extraction table but are not among the 13 families. The per-study extraction and the counting script are released with the companion repository, so that every bar in Fig. 9 can be regenerated from the published table.

**Limitations.** Studies were selected to represent each fusion paradigm, application, and evaluation practice, and works with quantitative evaluation and available code or data were preferred when several comparable studies existed; the resulting corpus therefore reflects the paradigms and applications that define current MMIF practice. Database coverage differs across venues, and records that could not be retrieved in full were assessed from the publisher record; both factors may exclude relevant studies.

### 1.4. Paper organization

The manuscript is organized as follows. Section 2 introduces imaging modalities and a three-axis taxonomy linking input composition, fusion stage, and output purpose. Section 3 reviews core fusion paradigms from preprocessing to data-level, feature-level, and decision-level fusion. Section 4 discusses emerging directions for clinical translation, including interpretable, generative, robust, and lightweight methods. Section 5 summarizes task-oriented clinical applications. Section 6 organizes open resources, evaluation metrics, and deployment tools for reproducible MMIF. Section 7 discusses challenges and future directions, and Section 8 presents the concluding remarks.

## 2. Foundations of Medical Image Fusion

**Modalities.** Medical image fusion is motivated by the complementarity of imaging modalities. MRI provides high soft-tissue contrast and, through sequences such as T1, T1ce, T2, and FLAIR, captures different anatomical and lesion-related characteristics [22, 23]. CT provides high-spatial-resolution anatomical information, especially for bone, calcification, and density-related structures [24, 25]. PET and SPECT provide functional or metabolic information through radioactive tracer uptake, making them useful for identifying physiologically active regions that may not be clear in anatomical images [26, 27]. Because no single modality can simultaneously provide complete structural, functional, and temporal information, MMIF aims to integrate complementary sources while reducing modality-specific limitations [1].

**Three-axis taxonomy.** We describe a fusion configuration by $\mathcal{C} = (\mathcal{I}, \mathcal{S}, \mathcal{O})$: input composition $\mathcal{I}$ specifies what is combined, fusion stage $\mathcal{S}$ specifies where information is integrated, and output purpose $\mathcal{O}$ specifies what the system produces. The taxonomy describes complementary properties without assuming statistical independence. A configuration can combine multiple input dimensions, fusion stages, and output purposes.

**Axis 1: input composition.** We record $\mathcal{I} = (\mathcal{M}, \mathcal{T}, \mathcal{A})$, where $\mathcal{M}$ identifies imaging modalities or sequences, $\mathcal{T}$ identifies explicitly integrated acquisition times or dynamic phases, and $\mathcal{A}$ identifies auxiliary guidance, if used. Multi-modality fusion combines distinct modalities (e.g., CT and MRI) or complementary MRI sequences within this review's scope. Multi-temporal fusion combines observations from different times, including dynamic phases within one examination or longitudinal visits. These properties can coexist: integrating several MRI sequences over follow-up visits is both multi-sequence and multi-temporal. Conversely, fusing multiple phases of one MRI sequence is temporal fusion without cross-modality fusion. MTFN illustrates this boundary by combining the first and third DCE-MRI phases to synthesize the eighth phase [28]; it is a related temporal synthesis example, not a core cross-modality MMIF method.

**Relationship to multi-source fusion.** Multi-source describes provenance, including modalities and acquisition times, rather than an additional fusion axis. Records, clinical tables, reports, and genomic measurements are outside the in vivo imaging scope of this review. Radiology–pathology fusion is an adjacent image-based extension outside our in vivo focus [29, 30]. Generated text may guide multiple-image fusion, as in MTG-Fusion [31]; multi-center acquisition alone does not constitute fusion.

**Axis 2: fusion stage.** The data, feature, and decision hierarchies describe the information granularity at the actual integration operation [32]. We record $\mathcal{S}$ as one or more of these stages, using the following assignment rules. Fig. 2 illustrates their architectural differences.

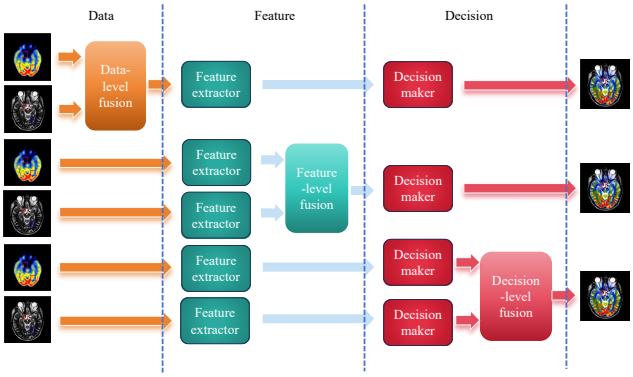



**Figure 2:** Architectural comparison of the three fusion hierarchies (example images: AANLIB [33]).

1) **Data-level fusion** combines raw pixels, voxels, or registered input channels before modality-specific representation extraction; direct intensity combination and raw-channel concatenation are examples [1, 6].

2) **Feature-level fusion** combines extracted representations, including transform sub-bands, handcrafted descriptors, or learned embeddings. Classification follows the representation being combined, even when a decoder subsequently reconstructs a fused image [19, 34].

3) **Decision-level fusion** combines modality-specific predictions or evidence over task labels, such as masks, probabilities, diagnostic scores, or belief masses. A shared classifier after feature fusion does not by itself constitute decision fusion [35, 3].

A multi-stage label is assigned only when distinct cross-modality integration operations occur at more than one stage. A sequence of an encoder, a single fusion module, and a classifier contains one fusion stage. The stage label identifies the integration operation; network family and output purpose are separate properties.

**Axis 3: output purpose.** We record the delivered output as a fused image, a segmentation mask, a detection result, or a classification/prediction, with multiple labels when applicable. Output purpose does not determine fusion stage: feature integration can generate a fused image or a segmentation mask, and segmentation can use early input concatenation, feature integration, or decision fusion. Table 2 maps concrete configurations across the axes and identifies a temporal boundary example. Together, the axes distinguish imaging inputs, integration operations, and delivered outputs.

In Table 2, "no explicit temporal fusion" means that acquisition time is not modeled as a fused input dimension; it does not imply simultaneous acquisitions. The raw-channel row is an illustrative configuration, while Zhang et al.'s proposed feature-fusion method is listed separately.

## 3. Fusion Paradigms: From Data to Decision

After establishing the basic modalities, dimensions, and hierarchies of MMIF, this section examines how fusion is implemented in practice. After preprocessing, fusion may occur at the data, feature, or decision stage, or at several explicitly identified integration points. These are alternative or complementary design choices, not mandatory sequential steps.

### 3.1. Fusion pipeline

Building on the taxonomy above, this section reviews how MMIF is technically performed. As shown in Fig. 3, a typical workflow includes data preprocessing, multi-level fusion, and downstream clinical application. Preprocessing reduces spatial and intensity heterogeneity; data-level fusion integrates raw inputs; feature-level fusion models cross-modality representations; and decision-level fusion combines semantic outputs for clinical inference [18, 1, 19]. Evaluation should match the intended output: fused-image quality, segmentation, detection, classification, or a defined clinical-use endpoint. Algorithmic performance alone does not establish clinical applicability.

### 3.2. Preprocessing

Preprocessing provides consistent inputs for fusion by reducing modality, spatial, and intensity differences [36, 37, 38, 39]. CT windowing and intensity scaling should retain the intended attenuation range; resampling and region extraction must preserve spatial correspondence. The chosen settings should be recorded with the fusion experiment. For MRI, N4ITK addresses intensity nonuniformity [40], while SynthStrip supplies a brain-extraction method [41]. Their roles should be distinguished from sequence-specific normalization or denoising, whose settings require separate specification. PET quantification depends on acquisition and correction procedures; attenuation and partial-volume correction address different sources of bias [42, 43]. PET standardized uptake values should not be assumed to apply to every SPECT dataset. Pseudocolor mapping is a display operation and should preserve access to the original quantitative image.

Spatial registration is a critical step because many fusion methods assume anatomical correspondence across modalities. Rigid registration is suitable for mild pose differences, whereas non-rigid registration is needed for tissue deformation and larger anatomical mismatch [38, 11]. In addition, 3D medical volumes are often resampled and converted into 2D or multi-planar slices for computationally efficient fusion, while channel adaptation harmonizes grayscale, multi-sequence, and pseudocolor inputs [44, 45, 2]. Augmentation protocols may include shared spatial transforms of paired images and labels or study-specific synthesis [6, 46, 47]. Robustness gains must be established under the evaluated perturbations rather than inferred from the use of augmentation alone.

Dynamic inputs and temporal correspondence. In 4D imaging, each 3D volume also has an acquisition time or physiological phase. Fusion therefore requires both spatial correspondence and a documented temporal matching rule; a common spatial grid alone is insufficient. Respiratory-gated PET/CT provides an acquisition-based example: Cheung et al. sort PET and CT data by respiratory phase and compare gated with ungated measurements in a prospective study of 16 patients with liver lesions [48]. The comparison evaluates respiratory motion management and PET quantification. Record gating signals, phase bins, sampling intervals and motion transforms, and check residual mismatch before combining inputs. Multi-phase synthesis within one MRI sequence remains a temporal boundary case.

### 3.3. Data-level fusion

Data-level fusion directly integrates raw pixel or voxel information and is therefore the most intuitive fusion paradigm. It is useful when modalities are well aligned and the goal is to preserve visible structural or functional cues, but it is highly sensitive to spatial misregistration, noise, and modality-specific intensity differences [1]. Representative studies are summarized in Table 3.

Traditional data-level methods mainly include spatial-domain fusion, transform-domain fusion, and fuzzy logic-based fusion. Spatial-domain methods perform direct pixel-level integration or color mapping and are computationally efficient, but they are fragile under non-rigid deformation [49, 50, 51, 52]. Transform-domain methods, such as DWT and DCT, decompose images into frequency components

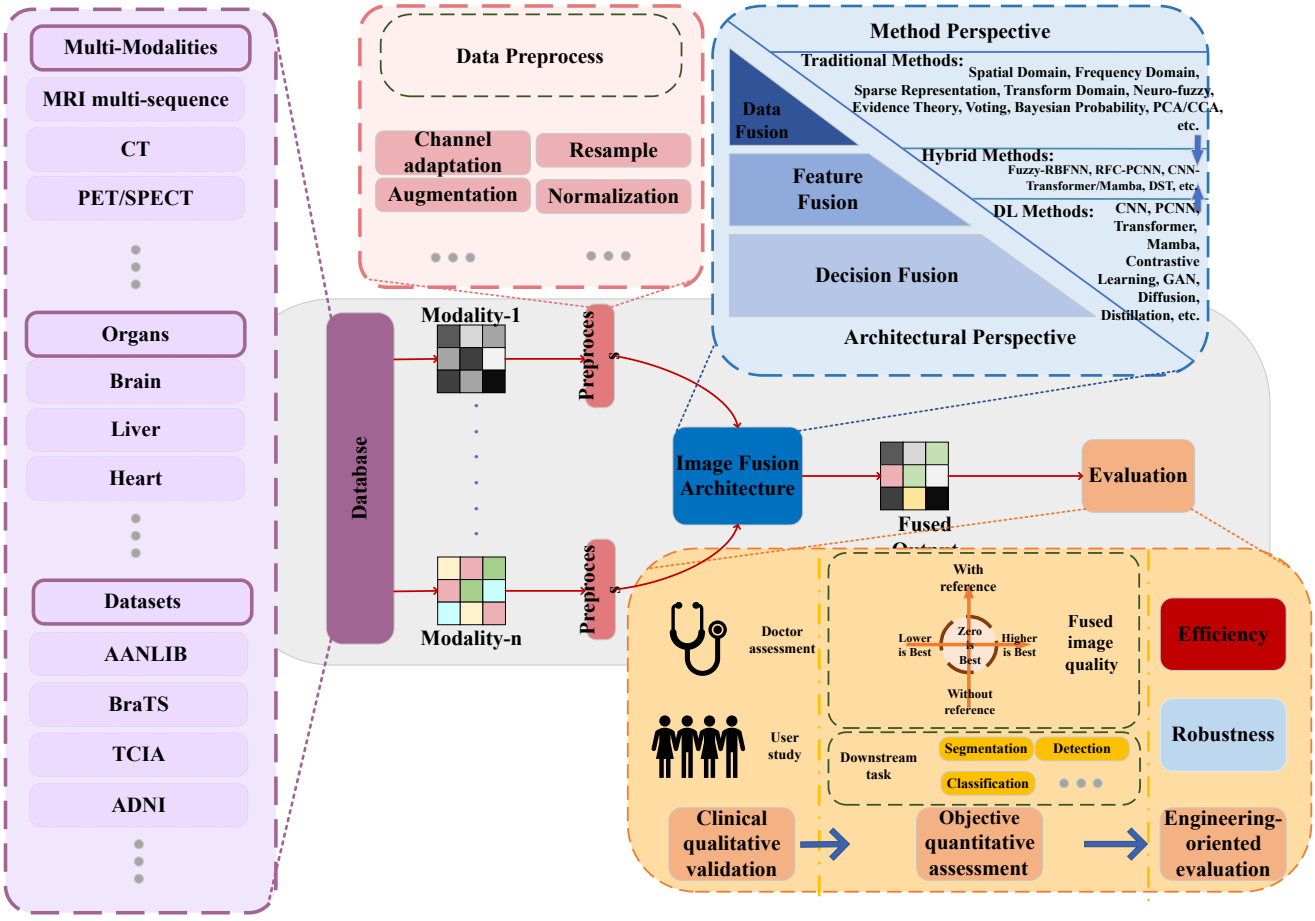



**Figure 3**: Overall pipeline of MMIF.

and define fusion rules for low- and high-frequency information, improving edge and contrast preservation compared with simple averaging [53]. Fuzzy logic-based methods model pixel ambiguity and intensity uncertainty, but their membership functions often require manual tuning and are difficult to scale across heterogeneous clinical datasets [54].

DL-driven data-level fusion includes early integration of raw images and learning-based spatial filtering. Modality synthesis is a supporting step when it supplies missing inputs to such a fusion workflow, rather than a fusion operation by itself. Early fusion stacks registered modalities as input channels and lets the network learn joint low-level patterns, but it still requires accurate alignment [55]. Modality-synthesis methods, such as PET synthesis from MRI or translation between MRI sequences, illustrate ways to complete imaging inputs [56, 57]. Their synthesis accuracy should be distinguished from the performance of any subsequent fusion step, and synthetic structures require validation. Hybrid filtering methods combine traceable pixel-domain operations with learned weighting or refinement, offering a compromise between interpretability and adaptability [58]. Data-level fusion remains a useful baseline when corresponding inputs are available and simple integration is adequate. Learned early fusion adds training requirements without automatically removing the dependence on registration; more elaborate representations should be justified by the target task and acquisition conditions.

### 3.4. Feature-level fusion

Feature-level fusion extracts modality-specific representations and then integrates them into a shared representation. It supports learned anatomical and functional complementarity, but feature extraction alone does not establish tolerance to spatial mismatch. Spatial feature combination still needs correspondence or an explicit alignment mechanism, as illustrated by BSAFusion [11]. Representative studies are summarized in Table 4.

Traditional feature-level methods rely on handcrafted feature extraction and fusion rules. Multi-scale transform methods, including DWT and NSST, decompose source images into sub-bands to preserve edges, textures, and structural details [59, 60]. Sparse representation methods map image patches or features into over-complete dictionaries and perform fusion in sparse coefficient space, improving noise robustness and spatial consistency [61, 62, 63, 64]. Their fusion rules are explicit, but computational cost varies: sparse coding requires repeated patch optimization, and dictionary construction may require offline learning [61]. Fixed

representations also require validation when acquisition protocols or anatomy differ from those used to select the rules.

DL-driven feature-level fusion learns representations and fusion rules jointly. CNN-based methods capture local structures and lesion boundaries through multi-branch encoders, attention modules, or spatial-frequency experts [65, 66]. Transformer-based methods model interactions within and across modalities [34, 67, 9]. Their cost depends on the attention design: global attention forms all-token interactions, whereas the shifted windows used in SwinFusion restrict attention computation to local windows. State space methods use selective scans for long-range modeling [68, 69, 70]. Linear scaling of a scan with sequence length does not by itself establish whole-model efficiency on high-resolution 3D medical data. Generative priors can support image fusion or modality synthesis [71, 72, 73]; their mechanisms and distinct output objectives are discussed together in Section 4.2. The main trend is toward hybrid architectures that combine local detail preservation, global dependency modeling, generative priors, and lightweight deployment.

SAM-style promptable segmentation connects an image encoder and prompt encoder to a mask decoder. MedSAM adapts this design to medical images using bounding-box prompts [74]; training across imaging modalities does not by itself implement fusion of several images of one patient. PCaSAM makes this integration explicit for T2-weighted, diffusion-weighted and apparent diffusion coefficient MRI inputs [75]. It freezes the MedSAM components while training a multimodal fusion module and an automatic bounding-box prompt generator. Cross-attention and multi-scale modality attention combine image embeddings before mask decoding. This feature-integration operation produces a segmentation. The study reports external-cohort testing with spatially aligned inputs, including registration where necessary; it does not establish performance for missing or unaligned sequences.

LLaVA-Med aligns biomedical visual and language representations and learns instruction following for image-based questions [76]. Such assistants are relevant to radiological interpretation, but image–text question answering alone is outside the detailed image–image comparison adopted here. Within that scope, MTG-Fusion uses generated descriptions to guide integration of multiple source images and produce a fused image [31]. Language thus supplies auxiliary guidance whose factual consistency with the sources requires checking. A generated answer, segmentation or diagnostic label does not alone identify decision-level fusion: that label requires integration of modality-specific predictions or evidence. Comparisons should separate the contribution of pretrained representations, prompt information and the actual fusion operation.

### 3.5. Decision-level fusion

Within the scope of this review, decision-level fusion combines modality-specific predictions or evidence derived from different medical imaging modalities or sequences. This stage integrates complementary imaging information after modality-specific prediction. Single-modality model ensembles lie outside this definition. Typical outputs include segmentation masks, diagnostic probabilities, lesion grades, or uncertainty estimates. This paradigm is useful when modalities are processed by specialized models or when clinical traceability is more important than fused-image visualization [77, 35]. Representative studies are summarized in Table 5.

Traditional decision-level fusion relies on probability rules, evidence theory, and voting strategies. Dempster–Shafer rules combine evidence from multiple medical images [77]. STAPLE estimates a consensus segmentation and rater-performance parameters from multiple segmentations [78]; it is a general label-combination method and qualifies as MMIF here only when the combined outputs originate from distinct imaging inputs. DL-driven decision fusion learns reliability and uncertainty from data. Deep evidential fusion estimates modality-specific confidence for tasks such as brain tumor segmentation and glaucoma grading [3, 79]. Cross-modal knowledge distillation is a supporting training strategy. Li et al. transfer information from an assistant modality to target-modality segmentation models [80]; the segmentation models use the target modality alone at inference. Reliability-aware decision fusion adjusts the contributions of imaging modalities according to their estimated confidence, but inaccurate reliability estimation may still degrade the fused prediction [35, 3]. For foundation-model systems, decision-level fusion likewise requires an identifiable combination of modality-specific predictions or evidence. Prompted segmentation or language generation alone is insufficient.

### 3.6. Cross-generation comparison and practical trade-offs

Fusion stage and model generation describe different properties: CNNs, transformers, state space models, and generative priors can serve different integration stages. Table 6 compares representative implementations across six practical conditions. These research directions overlap; all rows produce fused images except M3AE, which produces segmentation, precluding a single accuracy ranking.

**Registration and paired data.** Here, paired data means corresponding images of the same subject or scene, not a ground-truth fused image. Requirements must be stated separately for training and inference. MST-SR uses a dictionary learned from natural-image patches but fuses pre-registered source pairs [61]. EMFusion and SwinFusion learn source-preservation constraints from corresponding images without fused targets [2, 34]. DDFM reuses an unconditional pretrained prior without fusion-specific fine-tuning, yet still conditions inference on both source images [71]. Thus, removing fused-target supervision and removing paired-source dependence are distinct advances.

**Missing modalities and supervision.** Architectural capacity does not imply support for absent inputs. M3AE explicitly learns from modality masking and uses segmentation labels during fine-tuning; its reported experiments derive missing combinations from complete, co-registered MRI cases [12]. This relaxes input completeness at deployment under the evaluated protocol, while naturally incomplete training sets need separate assessment. Joint alignment addresses another constraint: BSAFusion learns from synthetic deformations of registered pairs and still requires both modalities [11]. Missing modalities, absent task labels, and misregistration should therefore be recorded independently.

**Cost and robustness.** Comparisons must separate offline learning from inference and account for sparse optimization, attention or scan operations, and generative training or sampling [61, 81, 34, 71, 70]. Table 6 records these cost sources. Timings from different hardware, resolutions, or implementations do not support a common efficiency ranking. Unequal-resolution, missing-sequence, and synthetic-deformation tests likewise establish different forms of robustness; none alone establishes reliability under combined perturbations or an unseen clinical site.

Task purpose constrains evaluation but does not uniquely select a fusion stage. For example, both SwinFusion and MTG-Fusion produce fused images through feature integration, whereas segmentation can use raw-channel concatenation, feature fusion, or evidential decision fusion (Table 2). Stage selection should therefore also consider registration quality, the representations available for integration, modality completeness, supervision, computational resources, and reliability requirements; the output task alone is insufficient.

**Reported quantitative comparison.** Table 7 consolidates the 16 methods evaluated in the GLIS-RT CT–T1-Gd MRI study of Safari et al. [8]. It reports fusion scores for all 16 methods and tumor-contouring Dice where available; unreported task results are marked NR. The source reports 2,276 test slices from a 230-patient collection; the number of test patients and the contouring subset are not specified. Parameters and FLOPs are also unreported. The comparison summarizes published results for this cohort. Differences between methods should be interpreted within its data and evaluation protocol; unreported cost measures prevent a common performance–cost comparison.

## 4. Emerging Directions for Clinical Translation

Beyond the core data-level, feature-level, and decision-level paradigms, recent MMIF research has increasingly focused on the requirements that determine whether fusion models can be translated into clinical workflows. This section therefore organizes emerging directions according to four clinical translation concerns. Interpretable fusion addresses the need for transparent and traceable fusion mechanisms. Generative fusion introduces image priors, with different requirements for source pairing, missing inputs, and supervision depending on the formulation. Robust fusion focuses on maintaining stable performance under noise, artifacts, degradation, and spatial misalignment. Lightweight fusion aims to reduce computational complexity and improve deployment feasibility in resource-constrained clinical environments. Together, these directions extend MMIF from algorithmic performance optimization toward clinically reliable, reproducible, and deployable medical image analysis.

### 4.1. Interpretable fusion

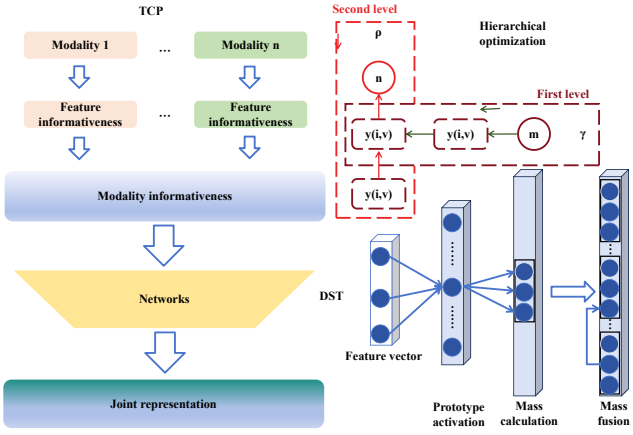



**Figure 4**: Representative interpretable MMIF studies.

Interpretability is essential for clinical translation because fused images or downstream predictions must be traceable to clinically meaningful modality contributions. As illustrated in Fig. 4, interpretable MMIF mainly develops along three routes: explicit feature disentanglement, probabilistic uncertainty modeling, and evidence-theoretic reliability learning. Feature disentanglement aims to separate modality-shared anatomical structures from modality-specific pathological or metabolic cues. Bilevel integrated optimization [82] provides a hierarchical mechanism for retaining structural content while suppressing noise, whereas CDDFuse [9] uses invertible networks to distinguish shared and private representations, making the source contribution of fused features more traceable.

Probabilistic methods further improve interpretability by assigning confidence or uncertainty to modality contributions. DDFM [71] introduces Bayesian estimation into diffusion-based fusion to adaptively balance anatomical and functional information, while uncertainty-weighted strategies for incomplete multi-modality segmentation show how reliability estimates can guide downstream decisions [83]. Evidence-theoretic methods, especially those based on Dempster-Shafer theory, are particularly relevant to clinical reasoning because they can model both confidence and ignorance. Contextual discounting and deep evidential fusion learn modality-specific reliability for segmentation tasks [35, 3], and CDE-Net further provides reliability-weighted pixel attribution maps that relate fused decisions to specific MRI sequences [84]. These studies illustrate how modality-specific reliability estimates can make image-based fusion decisions more traceable [3, 84].

**Trade-off.** Explicit decomposition exposes how image information is combined, whereas evidential decision fusion attaches learned reliability to task outputs. The latter requires task supervision and validation of the reliability estimates [3]. An attribution map or uncertainty score should therefore be assessed for faithfulness or calibration under the intended perturbations; its availability alone does not demonstrate robustness to missing or misregistered images.

**Source attribution.** Maps of modality contributions should link each highlighted region to the corresponding source slices and coordinates, including the registration transforms. Their faithfulness should be tested by controlled modality removal and local perturbations, checking whether changes in the selected output agree with the attributed contributions. Perturbation artifacts must be considered when interpreting these tests. Model- or training-label-randomization checks can further test whether learned explanations depend on the model and data, as motivated by saliency-map studies [85]. These diagnostic checks assess explanation faithfulness; clinical interpretation also requires source-image review and validation for the intended task.

## 4.2. Generative fusion

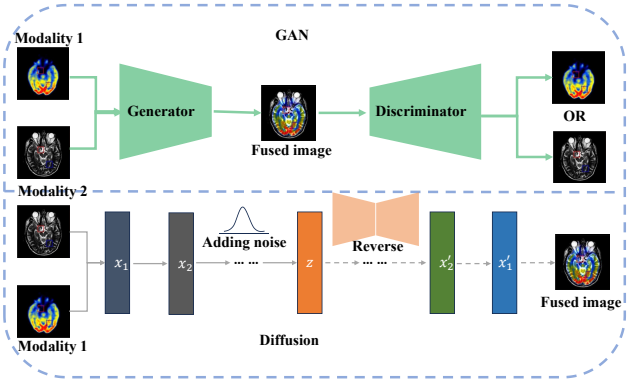



**Figure 5**: Common generative paradigms in MMIF (example images: AANLIB [33]).

Generative fusion uses learned image priors to construct outputs from complementary sources. Supporting modality synthesis has a different objective: estimating an absent imaging input before any subsequent fusion. Fig. 5 summarizes the main generative paradigms. GAN-based fusion learns source-preservation objectives through adversarial training, while modality translation uses a related architecture for a different output objective. Patch-level and spectral-normalized discriminators, such as U-Patch GAN [86], improve texture preservation, while dual-discriminator frameworks such as DDcGAN [81] separately constrain structural and functional information. Task-guided adversarial models, including AdvTCUnet [87], further connect generation with downstream segmentation objectives rather than optimizing visual fidelity alone.

Diffusion models offer another way to incorporate a learned image prior, with a different training and inference cost profile from adversarial models. DDFM [71] combines a pretrained unconditional diffusion prior with source-conditioned Bayesian inference without requiring a ground-truth fused target. DM-FNet [72] decouples diffusion-based feature reconstruction from the final fusion stage to reduce inference cost, while Mask-DiFuser [88] and DRMF [89] extend diffusion priors toward unaligned or degraded inputs. **Trade-off.** DDcGAN trains on paired medical patches with source-content and adversarial constraints; inference uses the generator alone [81]. DDFM reuses a prior without fusion-specific fine-tuning but requires iterative sampling [71]. Neither formulation removes dependence on both source images. Missing-modality completion requires a separate mechanism and validation; plausible generated detail can conceal unsupported structures and must be checked against observed anatomy [90].

## 4.3. Robust fusion

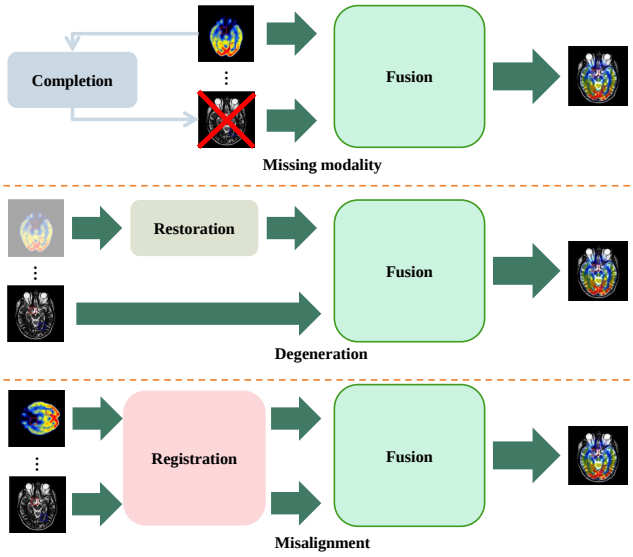



**Figure 6**: Representative robust MMIF studies for low-quality data (example images: AANLIB [33]).

Robust fusion targets the gap between ideal experimental assumptions and real clinical data, where images may be noisy, degraded, incomplete, or spatially misaligned. As shown in Fig. 6, recent work has moved from isolated restoration or registration modules toward joint optimization of degradation recovery, modality completion, alignment, and fusion. Within medical image fusion, UniFuse [38] illustrates the joint treatment of image restoration and fusion under non-ideal acquisition conditions. Joint optimization aims to reduce error propagation between processing stages, although strong out-of-distribution distortions and broad multi-center validation remain open issues.

Missing-modality robustness is another important clinical requirement, especially for multi-sequence MRI protocols. M3AE uses masked reconstruction followed by supervised segmentation and self-distillation, allowing one model to process the evaluated nonempty MRI subsets

[12]. MCF-Net instead uses complementary masking for supporting modality synthesis in distributed IoMT scenarios [91]; reconstruction and segmentation performance answer different questions. Spatial misalignment is addressed by joint registration-fusion frameworks such as BSAFusion [11], which decomposes large deformations into progressive alignment steps, and by diffusion-based approaches such as Mask-DiFuser [88], which learn to fuse substantially unaligned heterogeneous data. For degradation and artifact suppression, EFCNet [92] and DRMF [89] incorporate edge-aware or degradation-aware priors to preserve useful structural information while reducing noise amplification. **Trade-off.** Missing-input training, alignment, and restoration target different failure mechanisms. BSAFusion evaluates misalignment generated from registered medical pairs [11], whereas M3AE evaluates withheld MRI sequences on co-registered volumes [12]. These results support their respective tested conditions. Claims of broader clinical robustness require separate and combined perturbation tests, including real acquisition mismatch, naturally absent sequences, and external-site data.

**Fusion during intervention.** Intraoperative MRI–ultrasound fusion must update correspondence as tissue moves. Prada et al. describe fusion of preoperative MRI with intraoperative ultrasound for neuronavigation in 58 surgical patients, displaying an MRI slice coplanar with the ultrasound view and updating registration to compensate for brain shift [93]. The study provides alignment and feasibility evidence for a registration-supported fusion display; improved surgical outcomes require controlled clinical comparisons. The workflow links image coordinates to the patient and tracked probe, updates alignment from current intraoperative information, and preserves access to both sources. Deformation and resection can invalidate earlier correspondences. Motion compensation, temporal constraints and quality checks therefore need testing across procedural stages; temporal smoothness must not erase genuine anatomical change or conceal a failed update. Acquisition, tracking, registration, fusion and display all contribute to feedback delay.

### 4.4. Lightweight fusion

Lightweight fusion is motivated by the deployment requirements of MMIF, including real-time inference, limited memory, and cross-institutional generalization. Fig. 7 summarizes representative efficient designs. Neural architecture search (NAS) automates architecture design under task and hardware constraints. Early 3D NAS explored task-specific backbones [94], while latency-aware NAS [95] and ALMF-Net [96] introduced efficiency and cross-modality generalization into the search process. These methods reduce manual design effort, but searched architectures may still be brittle under severe degradation or misalignment.

Knowledge distillation compresses high-capacity fusion or segmentation models into lighter networks while preserving modality knowledge. Online mutual distillation

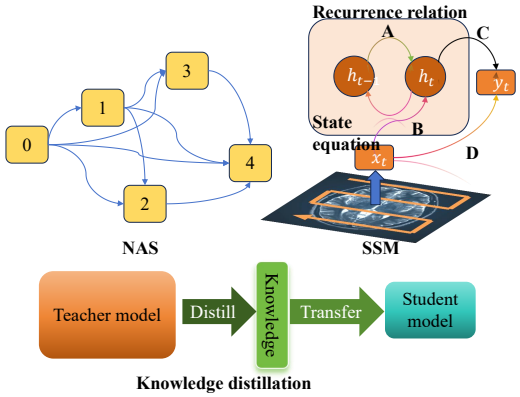



**Figure 7:** Representative lightweight MMIF studies (example images: AANLIB [33]).

[80] exchanges decision information between modalities, and DTMFusion [97] extends distillation to unaligned data by jointly optimizing deformation prediction and feature fusion. Efficient module design provides another route. CNN-Transformer hybrids such as TUFusion [39] and multi-receptive-field designs such as M4FNet [98] combine different spatial contexts; their computational benefit must be assessed against a specified baseline and input size. State space models such as Mamba offer selective scans whose cost scales linearly with sequence length for fixed model dimensions. DMCMFuse [68], MambaDFuse [70], and HCFMaNet [99] illustrate how cross-modality scanning and state-space modeling can support efficient fusion. **Trade-off.** Architecture search and teacher-based distillation may reduce deployment cost while increasing development or training cost. Selective scans reduce one source of sequence-processing cost, but channel width, scan directions, feature resolution, and implementation also determine memory and latency [70]. Efficiency comparisons should separate training from inference and report hardware, input dimensions, peak memory, and end-to-end latency including registration or completion. A smaller model also needs task and robustness checks after compression.

## 5. Task-Oriented Clinical Applications

This section considers applications in which information from multiple medical imaging modalities or sequences is integrated to generate a fused image or an image-based prediction. Method selection depends jointly on the output objective, acquisition conditions, available supervision, and deployment resources. Each application links representative studies to task-specific selection constraints.

### 5.1. Fused image generation

Fused image generation is the most direct application of MMIF. It aims to produce a single image that jointly preserves anatomical structures and functional information from complementary modalities. As shown in Fig. 8, this task is particularly useful when clinicians need intuitive visual interpretation, such as combining CT bone structures,

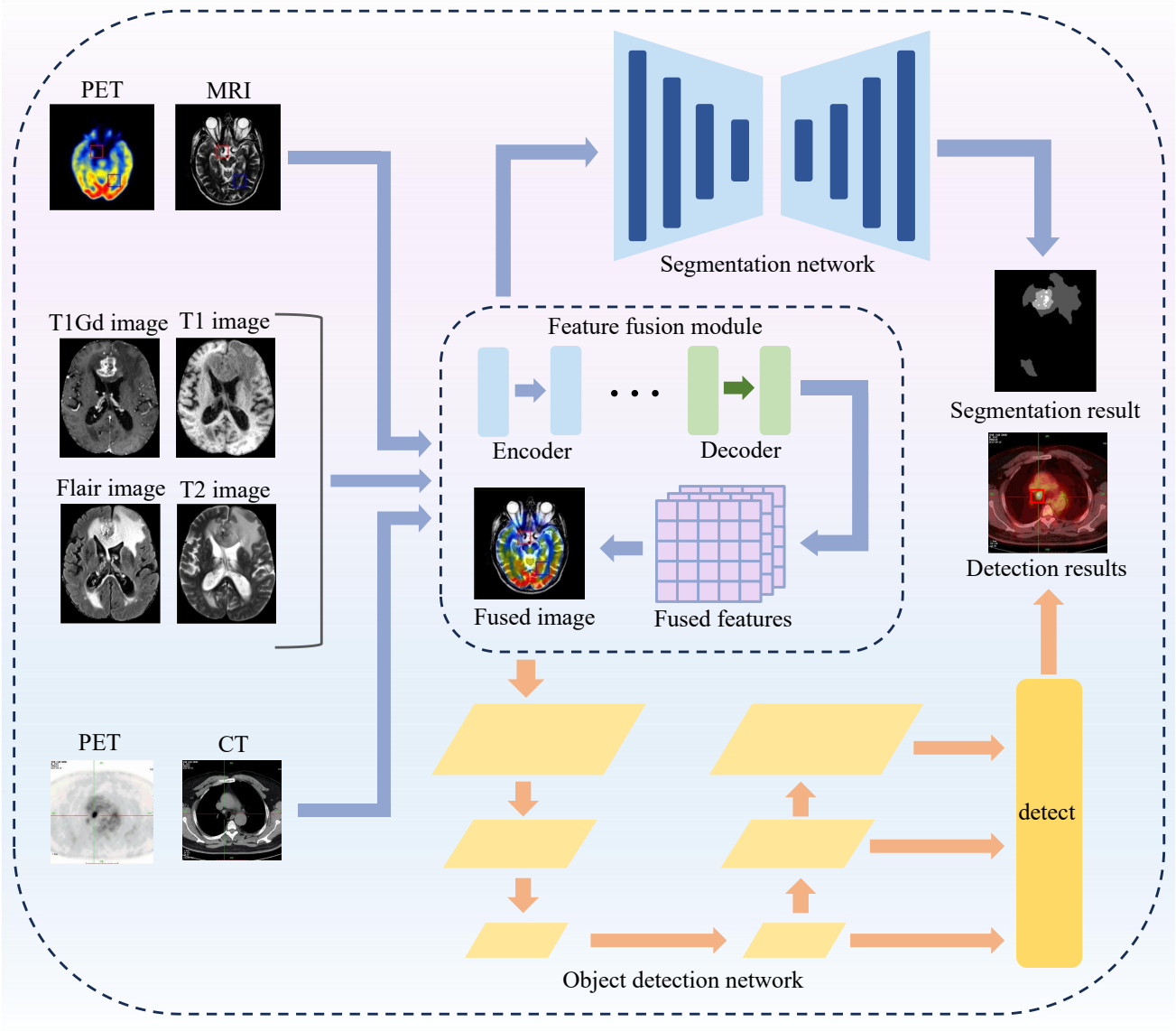



**Figure 8**: Common downstream applications in MMIF (example images: AANLIB for brain PET–MRI [33]; BraTS for brain glioma [22]; CHAOS for abdominal CT–MRI [100]).

MRI soft-tissue contrast, and PET/SPECT metabolic activity in one view [101, 102]. The central clinical motivation is not simply to improve visual quality, but to reduce information switching across modalities and help radiologists observe structural boundaries and functional abnormalities simultaneously.

This application is mainly associated with data-level and feature-level fusion, because the desired output is still an interpretable fused image. Existing studies mainly follow two routes. The first route is rule-driven or decomposition-based fusion, where source images are decomposed into base/detail layers or transformed into multi-scale representations before fusion. For example, Palanisami et al. used intuitionistic fuzzy sets with Gaussian-filter-based base/detail decomposition, obtaining fused images with improved contrast and fewer visual artifacts compared with several traditional baselines [103]. The second route is learning-based fusion, where CNNs, attention modules, transformers, or hybrid networks learn adaptive fusion rules from data. Fu et al. proposed MSRPAN to integrate multi-scale residual learning and pyramid attention, and validated it on MRI-CT, MRI-PET, and MRI-SPECT pairs from the Whole Brain Atlas, showing competitive objective metrics and fast inference [104]. Recent lightweight transformer-based methods further improve cross-modality interaction while targeting practical deployment constraints [105].

**Selection under constraints.** Complete, registered images and limited training data favor explicit transform rules as a baseline. Learned fusion requires source-preservation

objectives. Assessment should check subtle lesions, functional hotspots, over-smoothing, and color distortion. Evaluate alignment when inputs are misregistered and compare methods on the same modality pairs and reading task. Image scores alone do not establish diagnostic information preservation and should be interpreted alongside task and reader evidence.

### 5.2. Segmentation

Segmentation benefits from MMIF because lesion boundaries, edema, organ contours, and functional abnormal regions are often distributed across different modalities. In brain tumor analysis, for instance, T1ce highlights enhancing tumor regions, FLAIR captures peritumoral edema, and T2 provides complementary tissue contrast; relying on any single sequence may lead to incomplete subregion delineation [106]. Therefore, segmentation-oriented fusion mainly aims to improve boundary localization, region completeness, and robustness under ambiguous contrast.

Most segmentation studies adopt feature-level fusion. Multi-encoder architectures process different modalities with modality-specific branches and then integrate multi-scale representations in the decoder. Zhang et al. used four independent branches for T1, T1ce, T2, and FLAIR MRI and fused these features through residual learning, achieving improved brain tumor segmentation on BraTS 2021 [6]. Yan et al. designed MHFN for lumbar spine MR segmentation, where T1W and T2W-STIR features are integrated through an adaptive group fusion module, improving segmentation performance on a clinical lumbar dataset [107].

A second line of work focuses on task-oriented image-level or feature-level fusion, where the fusion process is optimized for segmentation rather than visual quality alone. Fuse4Seg and related studies show that fusion modules can improve segmentation when the fused representation is constrained by task objectives rather than only image-similarity metrics [108]. More recent methods further introduce uncertainty modeling, modality reliability estimation, and missing-modality learning, including deep evidential fusion and masked autoencoder-based representation learning, to handle ambiguous tumor boundaries and incomplete MRI sequences [3, 12].

**Selection under constraints.** Complete sequences and adequate labels permit supervised multi-encoder fusion [6]. Expected sequence absence calls for explicit subset training and testing, as in M3AE; evidential fusion estimates reliability when modality contributions are uncertain [12, 3]. Both require assessment of spatial correspondence and small or ambiguous tumor regions. Compare region and boundary accuracy by available modality subset. Fused-image scores apply only to image-producing systems and alone do not establish improved segmentation or delineation; Table 7 provides a study-specific numerical example.

### 5.3. Detection

Medical object detection aims to localize clinically significant targets, such as nodules, tumors, plaques, or suspicious lesion regions. Its performance depends strongly on whether the input representation contains complete and discriminative information [109, 110]. Compared with segmentation, detection usually places greater emphasis on lesion sensitivity, false-positive control, and region-level correspondence across modalities. MMIF is useful because structural modalities can provide anatomical localization, while functional or molecular modalities can highlight abnormal activity that may be visually subtle in a single modality.

Detection-oriented studies mainly use feature-level and region-level fusion, because candidate localization requires both discriminative semantics and spatial correspondence. An important requirement is to maintain region correspondence between imaging modalities so that complementary evidence refers to the same candidate lesion. Chen et al. developed CRFNet for cervical lesion detection from paired acetic-acid and iodine colposcopic images. Proposal shifting and corresponding channel attention align and integrate candidate-region features across the two imaging conditions [111]. For PET-CT detection, ALYOLOv5 introduces cross-feature fusion and adaptive attention into the YOLOv5 framework, illustrating how anatomical and metabolic cues can be jointly used for lesion localization [7].

**Selection under constraints.** Region fusion requires candidates to refer to the same lesion across modalities. Comparisons should test controlled offsets and lesion-size strata, report detection endpoints such as mAP and sensitivity at a specified false-positive burden, and include candidate generation and alignment in latency measurements. Limited task annotations remain a constraint: a label-free image-fusion objective does not supply lesion labels.

### 5.4. Classification

The classification studies considered here combine complementary medical imaging information, such as MRI and PET, to produce a diagnostic prediction [112]. A diagnostic label or score specifies the output purpose. A classifier may operate on a previously fused image; feature-level fusion combines modality-specific image representations, and decision-level fusion combines modality-specific predictions or evidence. Algarni et al. developed an automated diagnosis pipeline that combines MR and CT images through registration, histogram matching, wavelet/curvelet fusion, resolution enhancement, and CNN-based classification, achieving an optimal diagnostic accuracy of 90.9% on a dataset of approximately 370 images [113]. For neurodegenerative disease analysis, Fang et al. combine MRI and PET classifier outputs in an ensemble evaluated on ADNI [112]. Stability in deployed diagnostic workflows requires separate assessment.

**Selection under constraints.** Subject-level decision fusion can combine modality scores without voxelwise matching when they refer to the same patient and target label. It requires calibrated classifiers and a validated rule for unavailable modalities; spatial image or feature fusion retains stronger correspondence requirements. Patient-level splits, disease-class balance, and external validation stratified by modality availability and site are needed to assess inflated

accuracy in small or single-center cohorts. Endpoints should match the claimed use, such as diagnosis, subtype discrimination, prognosis, or treatment response.

**Task-aligned fusion.** Across the above applications, an important trend is the joint optimization of fusion with downstream clinical tasks. For registration and fusion, MURF constructs a mutually reinforcing framework in which alignment and fusion are optimized together for CT-MRI and PET-MRI scenarios [114], while BSAFusion embeds bidirectional stepwise alignment into a single-stage fusion pipeline for unaligned medical images [11]. For task-oriented image fusion, Fuse4Seg links image fusion with segmentation objectives [108], while GIFNet explores cross-task interaction to improve generalization across fusion scenarios [115]. However, joint optimization is not automatically beneficial. In a BMVC workshop study using LRRNet fusion and U-Net segmentation on BraTS-MEN 2023, Li et al. report convergence difficulties for their tested joint-loss configuration [116]. The finding concerns that configuration and highlights the need to examine interactions between fusion and segmentation losses. Task-aligned optimization is appropriate when task labels and a defined endpoint are available. Separate fusion and analysis stages remain useful when modular validation or reuse across tasks is required; both designs should be compared under the same data, annotation, and resource constraints. Fused-image generation, segmentation, detection, and classification should each be evaluated using suitable endpoints, while the fusion-stage label is assigned separately from the actual cross-modality integration operation. No one-to-one correspondence between a clinical task and a fusion stage is assumed.

## 6. Open Resources and Evaluation

Reproducing an MMIF study requires traceable connections between data selection, spatial preprocessing, the specified fusion method, evaluation and execution conditions. General-purpose and medical-image software provide components for these stages. PyTorch supports model implementation, Docker packages an execution environment, and 3D Slicer supports image inspection; using these tools alone does not establish reproducibility. Table 8 maps representative components to inputs, outputs and records to retain. For paired CT–MRI fusion, the reproducibility protocol specifies the records and checks required at each stage.

**Companion repository.** The curated resource collection is available at `https://github.com/CHamYE/MEDFusion/`. Its directories follow the structure of this review (`Technical_System`, `Extended_Techniques`, `Applications`, `Open_Source`, and `paper`), and each entry records the bibliographic reference, imaging modalities, fusion level, reported metrics, and links to code or data where available. The repository also hosts the per-study metric extraction used in Fig. 9 and is distributed under the MIT license. It is maintained by the author team, with Z. Ye as the designated maintainer; updates are released annually and after major medical-imaging conferences, community corrections are handled through issues and pull requests, and each release records the date on which the links were checked. The links cited in Tables 9 and 8 were verified on 29 September 2026.

### 6.1. Preprocessing tools

Preprocessing tools should be selected according to the source of heterogeneity that may affect fusion reliability. For spatial inconsistency, registration packages such as Elastix, ANTs, SPM, ITK, and SimpleITK support rigid, affine, and deformable alignment, while MONAI and TorchIO provide medical-image transforms that can be synchronized across modalities in DL pipelines. The central requirement is not merely accurate single-image registration, but preservation of cross-modality correspondence after resampling, interpolation, cropping, and augmentation.

ROI extraction, bias correction, normalization, and augmentation tools serve complementary roles. ITK-SNAP and 3D Slicer are commonly used for anatomical inspection, ROI editing, and annotation, whereas N4ITK is frequently adopted for MRI bias-field correction. Albumentations, TorchIO, and MONAI are useful for data augmentation, but spatial transforms must be applied consistently to all paired modalities. Intensity processing should remain modality-aware: excessive perturbation of PET/SPECT may distort quantitative metabolic information, whereas moderate contrast normalization is often acceptable for CT or MRI. For reproducibility, MMIF studies should report the registration strategy, interpolation method, voxel spacing, intensity normalization, ROI policy, and augmentation settings rather than only naming the software used.

### 6.2. Public datasets

Public datasets determine whether MMIF methods can be fairly compared across modalities, tasks, and clinical scenarios. Table 9 summarizes commonly used MMIF datasets. MONAI and TorchIO provide image-loading and transformation components for medical-image datasets. Input readers, subject identifiers, modality pairing and spatial metadata must be configured and checked explicitly; batching does not establish correspondence between images. The key requirement is that all modalities and labels remain synchronized throughout preprocessing, sampling, and augmentation.

In this review, AANLIB denotes the Harvard Whole Brain Atlas, the neuroimaging resource hosted at `https://www.med.harvard.edu/aanlib/`; both names identify the same resource. Fusion studies select MRI–CT, MRI–PET or MRI–SPECT examples from this atlas [117, 105, 118]. The selected cases and image pairs remain study-specific. BraTS provides a benchmark for brain-tumor segmentation from multimodal MRI [22]. Table 9 identifies the BraTS 2020 example and links its release description and challenge report; sequences, case counts and task scope must be tied to the selected release. TCIA hosts cancer imaging collections [119]; the TCGA glioma collections include curated MRI resources [120]. Collections should be selected according to the complementary images and task labels available for the intended fusion study. ADNI

includes longitudinal MRI and PET resources. Image-based fusion studies must define the selected imaging modalities, visits, participant overlap, and task labels; the official resource links are provided in Table 9. Dataset scale must be tied to a named release or challenge task, a counting unit, and a training/validation/test split. Participants, imaging visits, volumes, slices and selected image pairs are not interchangeable units. Table 9 therefore distinguishes fixed challenge releases from evolving repositories and longitudinal cohorts, with links to the corresponding sources (checked on 29 September 2026). The presence of several modalities in a resource does not establish complete paired inputs for every participant. For example, the CT and MRI cohorts in CHAOS contain different patients, whereas matching ADNI images requires subject- and visit-level reconciliation. TCIA collection totals must be distinguished from the subset with the complementary images and labels required for a particular fusion study. Harvard Whole Brain Atlas and legacy Imagefusion.org pair counts should be attributed to identified study subsets rather than treated as universal release sizes.

### 6.3. Training frameworks

The training framework should match the selected implementation, image dimensionality and execution environment. MATLAB supports prototyping with image processing functions, while transform-domain fusion and specialized fusion rules may require additional custom code. PyTorch and TensorFlow/Keras support custom architectures and losses; PyTorch Lightning organizes training loops and checkpointing. MONAI provides medical-image data and model components, and nnU-Net supplies a task-specific segmentation framework [44]. The study must still specify the fusion operation, modality/channel order, losses, training schedule, data split and checkpoint-selection rule. A framework name or saved weight file is insufficient to reconstruct these choices.

### 6.4. Evaluation and visualization

The evaluation of MMIF must prioritize clinical applicability, objective comparability, and engineering feasibility, because fusion is an intermediate process whose value cannot be determined by visual quality alone. A clinically meaningful evaluation protocol should therefore connect image-level fidelity, downstream task utility, and deployment feasibility rather than relying on isolated metric scores. Based on the surveyed studies, this review constructs a three-tier evaluation system that integrates qualitative clinical validation, objective quantitative assessment, and engineering-oriented evaluation, as illustrated in Table 10.

**Validation evidence and clinical applicability.** The metrics in Table 10 describe what is measured; study design determines what conclusions those measurements support. Tables 3–5 describe study design in the "Validation evidence" column. Its labels are descriptive and may be combined; they are not an ordinal evidence score or a certification of clinical readiness. Reporting the source population, study timing, and independent testing is consistent with CLAIM 2024 [121].

*Benchmark* denotes evaluation on public research datasets. *Retrospective* denotes an explicitly reported retrospective clinical cohort, with single- or multicenter scope where verified. *External* requires testing on an independent cohort excluded from training and model selection; using several datasets or centers alone does not meet this criterion. *Reader study* requires an identifiable reader group, assessment protocol, and endpoints, with qualifiers distinguishing clinician judgments from student ratings. Expert annotation and author-selected visual comparisons alone do not qualify. *Prospective* requires a prospectively conducted evaluation and must distinguish silent evaluation from use in a live clinical workflow; prospective data collection alone does not establish deployment. *Unclear* means that accessible primary information was insufficient to verify the relevant design, not that validation was absent. Thus, a public-dataset label or the word "Private" in the dataset column must not be interpreted as a complete description of clinical evidence.

These distinctions affect interpretation of individual studies. Deep evidential fusion combines public BraTS evaluation with a retrospective PET–CT cohort [3]. MAMS uses retrospective data from two hospitals, but the second hospital also contributes classifier training and validation; this does not establish independent external validation of the complete pipeline [122]. MRR includes radiologist ratings [123], whereas the methods section of LAGN reports a medical-student rating study [124]; neither image-rating result alone establishes improved diagnostic outcomes. Across these designs, strong image-quality or segmentation scores require further evidence of generalization, reader benefit, and workflow performance before clinical readiness can be claimed.

**Clinical Qualitative Validation** assesses reader judgments and perceived usefulness under a defined protocol, complementing quantitative evaluation. Reader evidence must identify who performed the assessment and what was measured. MRR reports radiologist ratings, whereas LAGN describes medical-student image ratings [123, 124]; these designs support different interpretations. Visual comparisons and annotated details provide illustrative evidence. Structured reader studies should report reader qualifications, blinding, comparator conditions, and prespecified endpoints, consistent with transparent study reporting [121]. Medical-student or other non-expert ratings can characterize comprehensibility, visual preference, or usability in the tested group; claims of diagnostic benefit require relevant clinical readers, a defined task, and diagnostic endpoints. Reading time and decision confidence should be assessed alongside accuracy and error rates, since faster reading or greater confidence alone does not demonstrate improved diagnosis. For example, the LAGN study reports image ratings, while the accessible EDPSN report does not provide sufficient reader-protocol detail for the same classification [124, 125].

**Failure detection.** MMIF systems that produce images should be evaluated for lesion omission, unsupported structures, and misleading color mappings. For *lesion omission*, use source-based lesion annotations and blinded paired reader comparisons to assess whether lesions and boundaries remain visible. Report lesion-level sensitivity or omission rate under a prespecified matching rule, stratified by lesion size and contrast, alongside global segmentation or image-quality scores. For *unsupported structures*, compare suspicious regions with each spatially corresponding source image, using local discrepancy or structural-consistency maps to identify candidates for expert adjudication. Report confirmed false structures per image or patient and detector sensitivity at a stated false-alert burden. Registration errors and genuine modality-specific signals must be checked before labeling a discrepancy as hallucination; direct cross-modality intensity subtraction alone is insufficient.

For *color-mapping failures*, verify how source intensities map to colors, their ordering where applicable, saturation or clipping, and hotspot location against the original functional image. Reader evaluation should assess whether the chosen color scale and blending opacity mask lesions or distort the perceived extent of uptake. The display should retain an explicit color legend and access to native PET/SPECT values and quantitative images, so that a rendered fusion color is not treated as a quantitative measurement.

**Uncertainty communication.** Region-level warnings should identify the affected source modalities and the basis of concern, such as registration quality, missing inputs, or a distribution-shift indicator. Predictive uncertainty, modality reliability, and input-quality flags should be identified separately. Probabilistic task outputs require calibration and risk–coverage evaluation. For image-only fusion, discrepancy maps, reconstruction residuals, or variation across generated samples are candidate warning signals; they require validation against defined failures and must not be presented as calibrated error probabilities. Usability assessment should test whether readers understand the warnings, recognize missed alerts, and can locate the original evidence.

**Out-of-distribution testing.** Keep independent hospital, scanner, acquisition-protocol, and disease-subgroup evaluations separate from controlled stress tests involving noise, motion, registration offsets, and missing modalities. Select alert thresholds on validation data and freeze them before testing. Report fusion or downstream performance, failure-detection sensitivity, false-alert burden, missed failures, and the proportion of cases referred for review in each setting. Synthetic perturbations help isolate failure mechanisms but do not substitute for external clinical data; an out-of-distribution score alone does not establish that a particular output is incorrect.

**Clinical validation and intended use.** A fusion output used for visualization, quantitative measurement, lesion detection, treatment planning or navigation supports different claims and risks. The IMDRF clinical-evaluation framework distinguishes valid clinical association, analytical validation and clinical validation [126]; MDCG 2020-1 describes the corresponding clinical-association, technical-performance and clinical-performance evidence for medical device software under the EU MDR [127]. Applied to MMIF, the intended use should connect the output to the relevant clinical question. Technical testing should address alignment, source fidelity, repeatability and defined failures. Clinical testing should establish performance for the intended population, users and setting; reader studies are appropriate when the claim concerns assisted interpretation, while measurement or navigation claims require their respective reference standards and endpoints. External-site testing and assessment of user errors should complement the failure analyses above. The precise evidence package depends on the claim and risk; these frameworks do not provide a universal SSIM or Dice threshold for clinical acceptability.

**Regulatory pathways and lifecycle evidence.** In the United States, the applicable pathway depends on device classification and intended use: eligible devices may use 510(k) substantial-equivalence review, novel low- or moderate-risk devices without a legally marketed predicate may use De Novo classification, and high-risk devices generally require premarket approval (PMA) [128]. In the European Union, software qualification and classification under the MDR, including the applicable Rule 11 criteria, determine the conformity-assessment route; notified-body assessment is required where applicable [129]. The fusion operation alone does not determine a device class. CE marking and FDA marketing authorization should not be inferred from research performance or DICOM compatibility. The evidence must address the complete intended system, including source-image access, failure handling and user interaction. Clinical evaluation should remain current during use and software changes, informed by post-market monitoring [127].

**Objective Quantitative Assessment** supports objective comparison and can be divided into direct assessment, which evaluates fused-image quality, and indirect assessment, which measures performance on a specified downstream task. Direct assessment is often organized by whether a reference image is used [1, 21, 130]. Source-referenced measures such as SSIM, MI, PSNR, and CC compare a fused image with specified source images; the sources are not ground-truth fused targets. EN, SF, and SD describe output statistics without source comparisons. SCD is the Sum of the Correlations of Differences and requires both source images, and is therefore source-referenced [131]. These direct metrics are useful for objective comparison, but their clinical relevance requires careful interpretation because they do not necessarily measure downstream utility. Indirect assessment therefore tests whether fusion improves performance on a specified downstream task under a defined comparison. For segmentation, Dice and HD95 play complementary roles: Dice quantifies the overlap between predicted and true lesion regions, whereas HD95 measures boundary error, which is

important for radiotherapy target delineation. For classification, ACC reports the fraction of correct predictions and can conceal poor minority-class performance. AUC assesses ranking across thresholds; it should be interpreted with class prevalence, sensitivity, specificity, and the intended decision threshold. For detection, mAP summarizes precision–recall performance under the specified class and matching definitions; diagnostic utility additionally requires evaluation in the intended reading or workflow setting. These metrics quantify task performance under the tested model, dataset, and reference standard. They do not alone establish diagnostic benefit, which requires evaluation of the intended clinical use and relevant readers or workflow.

**Engineering-Oriented Evaluation** centers on clinical deployment feasibility, encompassing efficiency and robustness metrics. Model size, inference time, and computational complexity characterize implementation cost. Clinical feasibility additionally requires end-to-end measurements covering image loading, preprocessing, inference, and output delivery, under the intended hardware and workflow conditions. Robustness claims should identify the tested condition: M3AE evaluates missing MRI sequences, BSA-Fusion evaluates alignment under synthetic deformations, and UniFuse addresses unaligned medical-image fusion [12, 11, 38]. These tests should not be conflated with external-site validation.

To clarify the evaluation landscape, this review analyzes the application frequency of metrics over the 192 studies catalogued in Tables 3–5, as shown in Fig. 9. Source-referenced and output-statistics measures dominate MMIF evaluation: MI is reported in 88 studies, $Q^{AB/F}$ in 71, and SSIM in 66, whereas task-level measures remain far less frequent (DSC in 25 and Acc in 15). This imbalance is the quantitative basis for the evaluation concerns discussed below. SSIM was developed for general image-quality assessment [132]; its interpretation in fusion depends on the chosen source references and the clinically relevant structures. However, relying on these metrics without considering their assumptions can lead to evaluation bias in medical scenarios [1]. Comparisons with structurally different source modalities require an explicit reference convention. For example, MRI and PET encode different structural and functional information; agreement with one source does not by itself establish retention of information from the other. Source-specific inspection should therefore accompany global scores.

A more critical issue is the risk of "metric deception" [133], especially for output-statistics metrics such as EN and SF. High EN or SF does not by itself establish useful information transfer. Evaluations in night-vision image fusion illustrate limitations of statistical image-quality measures [134]; this cross-domain evidence does not quantify hallucination rates in medical fusion. For medical fusion, lesion omission and unsupported detail require source-based and task-specific checks, as described above.

**Conditions for using MI and EN.** MI and EN may be reported as supplementary descriptive measures for fused images when methods are compared on the same cases with fixed intensity preprocessing, quantization and histogram or density estimation. For MI, identify each source reference and the aggregation rule; a high score can favor dependence on one source without establishing retention of clinically relevant information from both. EN describes output variability, which can also increase with noise or artifacts. Neither measure should alone rank clinical usefulness or establish fusion success. Interpret them alongside source-based artifact checks, structural measures and endpoints for the intended task.

This vulnerability highlights the need to complement global statistical metrics with measures of source-information transfer, such as the Sum of the Correlations of Differences (SCD) [131], and edge preservation, such as $Q^{AB/F}$ [135]. SCD measures the complementary information transferred from each source to the fused image; a higher value generally indicates stronger information transfer, but it does not directly identify artifacts or generative hallucinations. $Q^{AB/F}$ denotes edge-information transfer from source images $A$ and $B$ to fused image $F$. The measure assesses gradient magnitude and orientation, but does not by itself establish lesion-boundary accuracy.

Consequently, to guide the community toward a more objective, reasonable, and stable evaluation system, it is argued that the MMIF field must shift from "random metric stacking" to a **mutually constraining evaluation protocol**. Infrared–visible semantic-fusion studies illustrate that visual and task objectives can differ [136]; this is cross-domain evidence. The medical contouring comparison in Table 7 provides the bounded MMIF example. Report source fidelity, visual assessment, and task performance together.

**Evidence and limits.** In Table 7, MEF has higher reported SSIM than MedFusionGAN (0.81 versus 0.62), but lower contouring Dice (0.89 versus 0.96). IFCNN likewise has higher SSIM (0.78 versus 0.62), but lower contouring Dice (0.92 versus 0.96). HMSD has higher entropy (5.58 versus 5.20), but lower Dice (0.85 versus 0.96). These comparisons describe published method means and do not establish paired-case effects or a general correlation. Dice was obtained by semi-automatic level tracing, rather than a shared trained segmentation network. The source does not specify the matched case-level outcomes or identical aggregation subsets needed for correlation analysis. Several methods also lack Dice after qualitative exclusions or without a reported numerical value. The available summaries therefore cannot support patient-level correlations or pooled correlations across studies.

**Availability of paired outcomes.** Reported fusion and task metrics do not always refer to matched observations. DDFM reports Harvard fusion scores without a corresponding medical-task outcome [71]. MambaDFuse reports medical fusion and downstream detection on a separate infrared–visible dataset [70]. In Fuse4Seg, the principal fusion and segmentation comparisons contain different comparator sets [108]. These results cannot be pooled as matched (SSIM/EN, Dice) observations. Testing predictive association requires the same cases, modality

combination, fusion configuration and task protocol, with the analysis unit and uncertainty specified. Image-quality scores therefore need to be interpreted alongside task outcomes. These reports do not establish a general predictive relationship between image-quality and task-performance metrics.

**Four complementary evaluation aspects.** We organize evaluation around four questions: what source information is retained, whether structure is preserved, whether artifacts are introduced, and whether the intended task benefits. The four aspects address complementary failure modes and can overlap; statistical independence and mathematical orthogonality are not assumed. (1) *Information retention*: MI or SCD describes relationships with the sources; EN describes output variability and cannot alone demonstrate useful information transfer. (2) *Structural fidelity*: SSIM and edge-transfer measures such as $Q^{AB/F}$ assess specified similarities, without guaranteeing lesion-boundary preservation. (3) *Artifact sensitivity*: measures such as $N^{AB/F}$ assess defined distortions, but do not establish the absence of hallucinated pathology. (4) *Task performance*: DSC and HD95, for example, assess segmentation overlap and boundary error without establishing clinical efficacy. The aspects must be chosen for the output and intended claim; a model that produces only a prediction does not require a fused-image score. Implementation, reference convention and failure-case review remain necessary.

**Interpreting discordant evaluation results.** Evaluation endpoints should be chosen for the intended use before comparing methods. For fused-image generation, visual ratings, source-information preservation, and downstream performance address different questions; task-only predictions should be judged using endpoints appropriate to their outputs. Better appearance with worse task performance supports a visual-presentation benefit only. Improved task scores accompanied by expert-identified lesion omissions or unsupported structures indicate a task-specific gain with unresolved fidelity concerns. Expert preference without a clear task improvement supports preference or perceived usefulness, while a statistically nonsignificant difference does not establish equivalence. Expert judgments must distinguish aesthetic preference from suspected clinically relevant errors verified against source images and an appropriate reference standard.

**Investigating and reporting disagreement.** Compare the same held-out cases under defined comparator conditions and prespecified primary endpoints. Reader studies should specify expertise, blinding, and randomized or counterbalanced reading order. Examine discordant cases for registration errors, reference-standard limitations, lesion size, disease subgroup, and reader variability. Adjudicate suspected omissions or false structures against the original modalities, following the failure-detection checks above. Report effect estimates and confidence intervals, reader agreement where applicable, and all evaluation dimensions, including unfavorable results. These reporting requirements are consistent with CLAIM 2024 [121]. A favorable metric or preference rating should not override an unresolved clinically relevant error. When disagreement remains, limit the conclusion to the endpoint, reader group, and setting supported by the evidence.

**Dynamic and intraoperative evaluation.** Report spatial target registration error using landmarks independent of those used to estimate registration, motion- or phase-dependent errors, and the delay from acquisition to displayed output. For gated imaging, compare quantification and lesion depiction under the specified phase-matching protocol [48]. For intervention, assess correspondence before and after deformation or resection and report failed updates, tracking loss and recovery. Temporal consistency should be assessed alongside preservation of real changes. Geometric accuracy and low latency alone do not demonstrate clinical benefit; outcome claims need an appropriate clinical comparison.

### 6.4.1. *Evaluation metrics libraries*

scikit-image includes PSNR and SSIM functions. TorchMetrics supports image and downstream-task measures within tensor workflows. MATLAB Image Processing Toolbox also provides image-quality functions; specialized fusion measures may require separately identified implementations. Each reported result should specify the function and software version, reference-image convention, intensity range, masks, spatial window, channel handling and aggregation rule. A source modality used as a surrogate reference must be identified and does not constitute a ground-truth fused image. Custom fusion metrics should be archived with their parameters and verification examples; support for every fusion metric should not be inferred from a library name.

### 6.4.2. *Visualization tools*

3D Slicer supports volumetric visualization, linked slice views, segmentation overlays and inspection of registration. ParaView offers additional scientific visualization options. For an MMIF comparison, source and fused images should be displayed with recorded orientation, slice positions, intensity windows, color maps and overlay settings. These displays support inspection but do not themselves provide evidence of diagnostic benefit. TensorBoard or an experiment-tracking service can retain curves and image summaries; exported logs should be linked to the data manifest, configuration and model checkpoint so that the displayed result can be traced to its inputs.

## 6.5. Deployment tools

Deployment components package and execute a specified MMIF pipeline. ONNX supports model interchange when the required operations are compatible with the target runtime; TensorRT and OpenVINO offer inference optimization on supported hardware. Export and reduced-precision execution can change numerical outputs, so comparisons with the source implementation

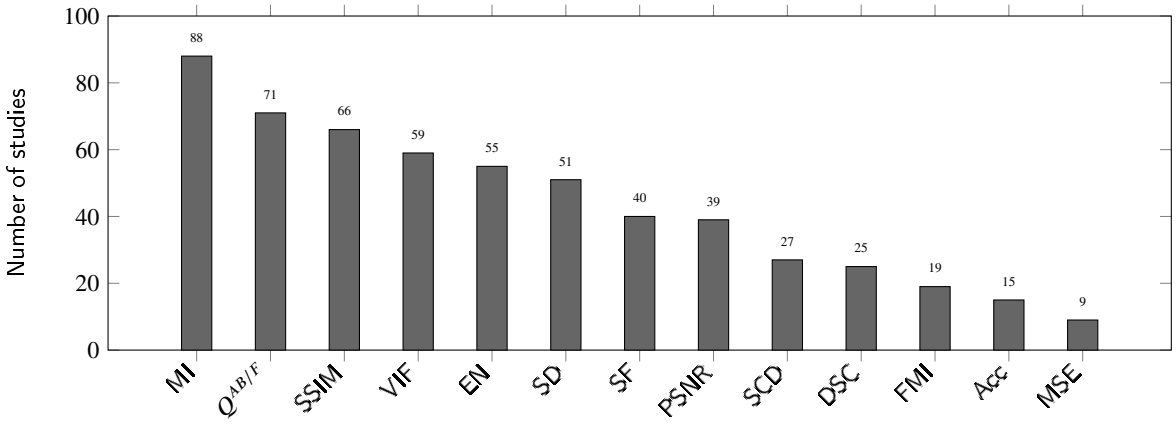



**Figure 9**: Application frequency of evaluation metrics in MMIF studies (192 studies catalogued in Tables 3–5; one count per study and metric family).

should use fixed inputs and documented acceptance tolerances [137]. Latency and memory reporting should include preprocessing and postprocessing, alongside model inference.

MONAI Deploy App SDK can connect configured image readers, inference operators and output writers. The application must define input-series selection, metadata handling and output semantics [138, 139]. Docker Engine packages software dependencies, but the build recipe, dependency versions, image digest, run command and host/GPU conditions must also be retained. Neither containerization nor DICOM compatibility establishes clinical readiness. Table 8 summarizes component roles and the records needed to reconstruct their use; the linked documentation and license information were checked on 29 September 2026.

### 6.6. Reproducibility protocol for paired-image fusion

For paired CT–MRI image generation, a reproducibility protocol should connect a data manifest to spatial preparation, fusion inference, evaluation and an archived run record. The selected fusion implementation remains a study-specific component. Segmentation or other task-prediction studies additionally require task labels, a defined task model and an evaluation protocol suited to those outputs.

**Data provenance and splits.** Record the release, source identifiers, paired files, exclusions and access conditions. Split by patient before extracting slices or patches, and keep all derived samples from a patient in the same split. If patient identifiers are unavailable, state that patient-level separation cannot be fully verified. Where images cannot be redistributed, provide permitted identifiers and acquisition instructions so that eligible users can reconstruct the subset.

**Spatial preparation.** Check the correspondence of the CT and MRI acquisitions, select the reference grid, estimate or verify registration, and retain the transform direction, spacing, orientation, interpolation, crop and normalization settings. Apply shared spatial transforms to paired images and labels, with interpolation appropriate to each data type. Record visual or quantitative registration checks and exclusions. Pass the prepared inputs in a fixed modality order, preserving the mapping back to the original images.

**Execution and evaluation.** Archive the code revision, dependency versions, model configuration, weight-file hash, run commands, seeds, deterministic settings and hardware. For retraining, also provide losses, optimizer, schedule and checkpoint-selection rule. Evaluate identical held-out pairs with fixed metric implementations, reference conventions, intensity ranges and aggregation rules; save per-case results, failures and runtime measurements. Exact numerical agreement across platforms should not be assumed from a shared seed or container.

**Run archive and verification.** Link the manifest, preprocessing parameters, configuration, checkpoint and metric script under one run identifier. Retain fused outputs, logs, permitted example inputs and expected outputs, with a stated numerical tolerance. A repeat run in a fresh environment should check input counts and identifiers, image geometry, output agreement and metric summaries. Report whether only inference was repeated or training was also reproduced, together with any deviations. A reproducibility record therefore links the software environment to the selected data, configuration, outputs and verification results.

## 7. Challenges and Future Directions

The remaining challenges of MMIF can be understood as limitations along the same translation chain developed throughout this review: heterogeneous data acquisition, reliable fusion modeling, clinically meaningful evaluation, reproducible implementation, and safe deployment. Fig. 10 summarizes the main challenge groups and their corresponding research directions.

### 7.1. Core challenges

The reviewed method conditions and task constraints reveal the following cross-task gaps for clinical translation.

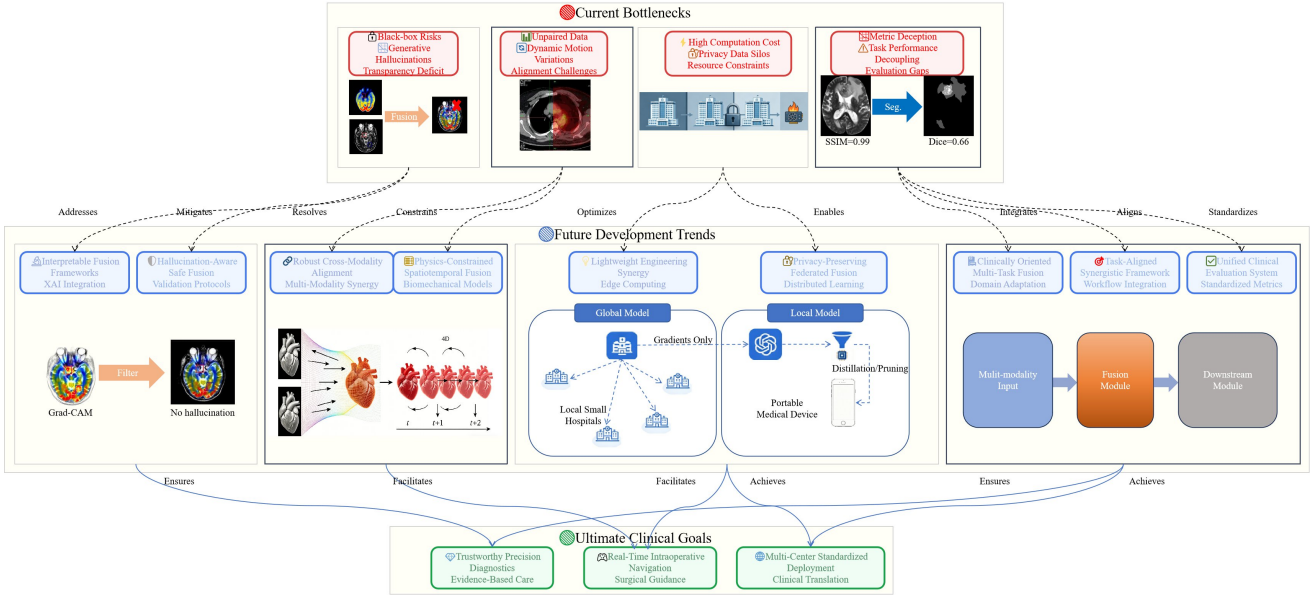



**Figure 10**: Future trends and challenges in MMIF.

**Combined acquisition shifts.** Alignment and representation methods address specific sources of modality heterogeneity [9, 117, 34, 140]. Their reliability when missing inputs, misalignment, noise, and scanner or protocol changes occur together remains insufficiently established [91, 38]. This gap extends beyond testing each perturbation separately.

**Interpretable clinical evidence.** Attention, uncertainty, and evidence-theoretic mechanisms require a validated link between modality contributions and clinical findings [3, 84, 79]. The remaining question is whether such explanations support reliable interpretation and intervention by clinical users.

**Comparable task and clinical evidence.** The GLIS-RT comparison in Table 7 illustrates disagreement between image-quality and contouring outcomes within one study [8]. Inconsistent metric implementations, preprocessing, and clinical validation limit comparisons across studies [1, 62, 21].

**Deployment evidence.** Resource-efficient architectures require end-to-end assessment. DICOM specifies medical-image information and communication conventions; MONAI Deploy supplies configurable readers, inference operators, and writers [139, 138]. These capabilities do not establish PACS interoperability for a particular application or sustained clinical performance.

**Privacy, safety, and dynamic imaging.** Privacy constraints and institutional data silos complicate multi-center validation [141, 142, 143]. Artifacts and missed structures documented in image reconstruction motivate MMIF-specific stress tests [144]; failure rates need to be measured for each fusion method and use condition. Dynamic and intraoperative imaging require consistent correspondence despite motion and anatomical change.

**Traceability and human override.** MMIF workflows should preserve original images and link fused results to source series, slices, spatial transforms, preprocessing, model version, and display settings. Clinical users should have synchronized source views, adjustable or removable overlays, and controls to reject outputs and request re-registration or reprocessing. Quality or uncertainty alerts should identify the affected region and reason, while source access remains available even without an alert because failures may be missed. Log interventions and subsequent processing. Reader and workflow studies should assess appropriate intervention, alert burden, and clinical safety.

## 7.2. Future directions

Five research questions emerge from the limitations of current fusion methods and their clinical evaluation.

**Calibrating confidence under changing input conditions.** Evidential fusion provides modality reliability estimates and predictive uncertainty [3, 84], but their availability alone does not establish calibration under acquisition or population shifts. Can confidence remain consistent with observed errors when a modality is absent, noisy, or acquired at another center? Candidate approaches include reliability-aware modality weighting, held-out calibration, and referral of uncertain cases for review; temperature scaling provides a baseline to test, rather than an assumed solution for MMIF [145]. For classification or segmentation outputs with probabilistic predictions, validation should separate calibration and test sets and report reliability curves, Brier score or negative log-likelihood, and risk–coverage curves alongside task performance. Missing-input, noise, and external-center tests should establish when confidence becomes unreliable. Image-only fusion requires a separately defined error or clinical endpoint before such calibration measures can be interpreted.

**Learning from clinically incomplete modality sets.** M3AE demonstrates missing-modality segmentation using simulated modality removal from complete, co-registered MRI data [12]. A remaining question is whether this robustness transfers to clinical cohorts in which acquisition depends on patient condition, cost, or examination protocol. Candidate methods include training across modality subsets, explicitly modeling acquisition patterns, and weighting available evidence by reliability. Direct fusion of incomplete inputs should be compared with completion followed by fusion under the same available information. Validation should distinguish incomplete training data from missing inputs at inference, include genuinely incomplete cohorts, and report performance for each available modality combination and the worst-performing subset. Registration requirements, missingness mechanisms, and subgroup composition should be documented so that simulated dropout results are not treated as evidence for every clinical missingness pattern.

**Adapting foundation models for joint imaging inputs.** PCaSAM provides evidence of foundation-model adaptation to joint MRI inputs under aligned-input conditions [75]. A remaining question is when such adaptation transfers to other cohorts or tasks, particularly with missing sequences or imperfect alignment. Frozen encoders with fusion modules, parameter-efficient adapters, and full fine-tuning should be compared with specialist models under matched training data, patient splits, and annotation budgets. Validation should disclose pretraining overlap and prompt information or interaction effort, test external centers and input perturbations, and report computational cost. Gains should be attributed separately to pretraining, adaptation, and multimodal integration through controlled ablations.

**Improving task utility while preserving lesion evidence.** Task-guided fusion links image construction to segmentation [108, 146], while diffusion-based fusion introduces generative priors [71]. These mechanisms motivate a specific question: can task performance improve without suppressing subtle lesions or introducing unsupported structures? Candidate approaches include source-consistency constraints, lesion-sensitive objectives, artifact detection, and traceable access to source images. Validation should compare joint training with modular fusion–analysis pipelines under matched data and resource conditions. For image-producing systems, lesion retention and false structures should be assessed alongside downstream performance, using lesion annotations and blinded reader assessment where feasible. Evaluation across lesion sizes, disease subgroups, and a second downstream task can reveal whether gains depend on selective information loss. These evaluations should identify which failure modes occur for a given method and use setting.

**Reducing workflow cost without losing reliability.** Compact networks and efficient sequence modeling address computational cost [105, 70], but parameter counts or scan complexity do not determine complete clinical workflow latency. How much can a system be compressed before task accuracy, calibration, or robustness deteriorates? Distillation, quantization, and hardware-aware optimization are candidate methods, informed by existing lightweight fusion and architecture-search approaches [96, 95]. Validation should compare the original and compressed systems on identical hardware, input dimensions, precision settings, and batch sizes. End-to-end latency and peak memory should include preprocessing, registration, any modality completion, fusion, and downstream inference. The same external-center and missing-input tests should be repeated after compression, with calibration assessed for probabilistic task outputs, to identify a usable accuracy–cost trade-off.

Privacy and dynamic imaging impose additional conditions on these questions. Federated evaluation should retain separate center-level results to expose heterogeneity [143]; any differential-privacy claim should specify its privacy budget and utility cost. For dynamic and intraoperative settings, research should examine whether motion models and temporal constraints remain valid through actual anatomical change. Across all five questions, shared patient-level splits, documented preprocessing and metric implementations, and reproducible code are needed to make validation comparable.

## 8. Conclusion

This review presents a systematic analysis of MMIF, covering its technical evolution, methodological framework, open resources, evaluation strategies, and application scenarios. Within the defined focus on in vivo medical imaging, the review relates imaging inputs and output objectives to data-level, feature-level, and decision-level fusion. It traces the evolution from traditional multi-scale transformation methods to DL-driven approaches for fused-image generation and image-based clinical prediction. The review also discusses extended directions that address practical challenges, including interpretable fusion, generative model-driven fusion, low-quality data-adaptive fusion, and lightweight generalized architectures. In addition, it summarizes an open-resource workflow that connects data preprocessing, model development, evaluation, visualization, and deployment, with the aim of improving reproducibility and supporting both algorithmic innovation and engineering implementation. By linking technical mechanisms to downstream application scenarios, the review clarifies how different fusion hierarchies correspond to practical clinical needs. Although substantial challenges remain, the future directions identified here provide a roadmap for subsequent research. MMIF integrates complementary imaging evidence into fused images and task-specific predictions; the broader integration of images with clinical records, reports, or omics remains a related field beyond this review's detailed scope. Its future development will depend on the joint refinement of technical methods and practical application constraints, supporting more accurate and efficient medical image analysis.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have potentially influenced the work reported in this review.

## Data availability

The material analyzed in this review consists of information and reported results from the cited publications and public resources. Study summaries and the reported quantitative comparison are provided in the manuscript tables. Curated links to publications, datasets and software are available at https://github.com/CHamYE/MEDFusion/. Access to the underlying imaging datasets and third-party code is governed by their original providers.

## CRediT authorship contribution statement

**Junxin Chen:** Conceptualization, Methodology, Writing–original draft, Writing–review and editing. **Zhiheng Ye:** Methodology, Software, Validation, Formal analysis, Writing–original draft. **Mingzhi Wang:** Investigation, Data curation, Formal analysis, Validation, Visualization, Writing–review and editing. **Jiawei Zhang:** Investigation, Resources, Visualization, Writing–review and editing. **Jun Mou:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing–review and editing. **Xu Xu:** Methodology, Validation, Writing–review and editing. **Yongfei Wu:** Formal analysis, Validation, Writing–review and editing. **Wei Wang:** Supervision, Writing–review and editing.

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# A. Supplementary table

**Table 1**
Topic coverage of representative medical image fusion reviews. The extraction criteria are defined in Section 1.2.

| Work | Year | Modality features | Public datasets | Prepro-cessing | Fusion methods | Metrics | Open tools/code | Challenges or trends | Task scope |
|---|---|---|---|---|---|---|---|---|---|
| [13] | 2014 | ✓ | × | ✓ | T | × | × | ✓ | Organ-based applications |
| [147] | 2015 | × | × | × | T | × | × | ✓ | Clinical image analysis |
| [148] | 2018 | ✓ | × | ✓ | T | ✓ | × | ✓ | Fused-image generation |
| [18] | 2019 | ✓ | ✓ | ✓ | D | ✓ | ✓ | ✓ | Segmentation |
| [16] | 2020 | ✓ | ✓ | × | T/D | ✓ | × | ✓ | Fused-image generation |
| [14] | 2020 | ✓ | × | ✓ | T | ✓ | × | ✓ | Fused-image generation |
| [1] | 2022 | ✓ | ✓ | ✓ | T/D | ✓ | ✓ | ✓ | General MMIF |
| [20] | 2023 | ✓ | ✓ | × | T/D | ✓ | ✓ | ✓ | Fused-image generation |
| [15] | 2023 | ✓ | × | ✓ | T/D | ✓ | × | ✓ | Fused-image generation |
| [17] | 2024 | ✓ | ✓ | × | T/D | ✓ | × | ✓ | Fused-image generation |
| [19] | 2024 | ✓ | ✓ | ✓ | D | ✓ | ✓ | ✓ | Classification |
| [21] | 2025 | ✓ | × | × | T/D | ✓ | × | ✓ | Diagnosis and treatment |
| This review | 2026 | ✓ | ✓ | ✓ | T/D | ✓ | ✓ | ✓ | Image and task outputs |

**Table 2**
Mapping representative configurations across input composition, fusion stage, and output purpose.

| Example | Input composition | Fusion stage and operation | Output purpose |
|---|---|---|---|
| Raw-channel concatenation (illustrative configuration) [6] | Multi-sequence MRI; no explicit temporal fusion; no auxiliary text | Data: concatenate raw input channels before a shared encoder | Segmentation |
| SwinFusion [34] | CT-MRI or PET-MRI; no explicit temporal fusion; no auxiliary text | Feature: cross-attention integrates image representations | Fused image |
| Deep residual feature fusion [6] | T1, T1ce, T2, FLAIR MRI; no explicit temporal fusion; no auxiliary text | Feature: combine modality-specific residual features | Brain tumor segmentation |
| Deep evidential fusion [3] | PET-CT or multi-sequence MRI; no explicit temporal fusion; no auxiliary text | Decision: combine reliability-discounted modality-specific belief masses | Tumor segmentation and uncertainty |
| MTG-Fusion [31] | Multiple medical images (e.g., CT-MRI); no explicit temporal fusion; generated text guidance | Feature: text-guided extraction and weighting of image features | Fused image |
| MTFN (temporal boundary example) [28] | One DCE-MRI sequence, phases 1 and 3 of one examination; no auxiliary text | Feature: co-attention integrates temporal representations | Synthesis of phase 8; related supporting task |

Table 3: Representative data-level fusion studies. Validation evidence is defined in Section 6.4.

| Paper | Year | Modalities | Method | Datasets | Metrics | Validation evidence |
|---|---|---|---|---|---|---|
| Fusion system[49] | 1993 | CT-MR | Color channel mapping | Private | Point displacement error, spatial dimension accuracy, visual evaluation | Unclear |
| 3D voxel fusion[50] | 2004 | CT-MRI, MRI multi-sequences | 3D reconstruction, color coding | Private | Fusion accuracy, FPS | Unclear |
| 3D spinal fusion[51] | 2004 | 3D CT-MRI | Registration, voxel merging, Semi-Boundary | Private | Visual evaluation | Unclear |
| Color table fusion[149] | 2004 | CT-PET-MRI | Anatomy specific color fusion, Observer oriented fusion | Private | Visual evaluation | Unclear |
| Regis+ICP[150] | 2007 | DE-MRI&PV-MRA, PV-MRA&EAM, DE-MRI&PV-MRA&EAM | Intensity-based rigid registration, ICP | Private | Registration error | Unclear |
| TRUS-MRI fusion[52] | 2007 | TRUS-MRI | Manual registration method for 3D US reconstruction | Private | Hausdorff, TRE, FRE | Unclear |
| AWT+curvelet transform[151] | 2010 | MRI-CT | Curvelet transform | Private | Similarity, PSNR | Unclear |
| PCA+DWT[152] | 2013 | CT-PET | PCA, DWT | Private | MSE, PSNR, Entropy, visual evaluation | Unclear |
| Improved ICP[153] | 2014 | CT& Blue-Ray | ICP, improved Voronoi | Private | Registration average Euclidean distance | Unclear |
| DCxWT [154] | 2014 | CT-MRI, T1w MRI-MRA | DCxWT, multi-scale max fusion | **Imagefusion.org** | Entropy, $Q^F_{AB}$, FF, SD, FS | Unclear |
| Block PCNN[155] | 2015 | CT-MRI | S-PCNN, OFG, Image block processing | Private | FD, MI, $Q^F_{AB}$, MV, RMSE | Unclear |
| QC hybrid[156] | 2015 | CT-MRI | Pseudo conformal mapping, mixed criteria | Private | $e_{max}/e_{mean}$, Beltrami, MI, visual evaluation | Unclear |
| CMA-ES[157] | 2016 | CT-MRI | Morphological filtering, evolutionary optimization | Private | Registration error, time, visual evaluation | Unclear |
| Union LAP[158] | 2016 | CT-MRI, PET-MRI, SPECT-MRI | Laplacian pyramid multi-scale decomposition, multi feature fusion, image statistics | **AANLIB** | PSNR, STD, SSIM, TMQI | Benchmark |
| QPSO-PCNN [159] | 2016 | CT-MR, MR T1-FDG, MR T2-SPET | QPSO-PCNN | Private | STD, SF, MI, SSIM, Sharpness | Unclear |
| Wavelet channel fusion[160] | 2017 | MR T1-CT, MR DWI-CT | Wavelet-Pseudo color-$\alpha$ channel | Human Connectome Project | IE, MI, MG, SF, visual evaluation | Unclear |
| MRR[123] | 2017 | CT-MRI | B-spline deformable registration | Kasturba Hospital | MI, visual evaluation | Reader study (radiologist ratings) |
| PCA-DWT[161] | 2017 | CT-MRI, MRI-SPECT | PCA, DWT, PCNN | **AANLIB** | PSNR, STD, Entropy, MI, $Q^F_{AB}$, visual evaluation | Unclear |
| LR sparse dictionary fusion[162] | 2018 | CT-MRI, MRI-T1/MRI-T2, MRI-PET, SPECT-MRI | Discriminative low-rank sparse dictionary learning, low-rank and sparse component decomposition, weighted nuclear norm regularization, max-absolute component fusion | **AANLIB**, **imagefusion.org**, private training samples | MI, $Q_{TE}$, $Q_{NCIE}$, $Q_M$, $Q_P$, $Q_S$, $Q_G$ | Unclear |
| CNN Type-I (input)[32] | 2018 | PET-CT-MRI T1/T2 | 2D patch stacking across modalities; channel-spanning kernels | **TCIA** | Pixel-wise Accuracy, visual evaluation | Benchmark |
| Fuzzy logic[54] | 2018 | CT-MRI-PD | T-S fuzzy reasoning system | **AANLIB** | MSE, PSNR, SNR, visual evaluation | Unclear |
| Tetrolet SUnSAL [63] | 2018 | CT-MRI, PET-MRI, SPECT-MRI | Tetrolet transform, SUnSAL, cycle-spinning, MDC, WSML | **Imagefusion.org**, **AANLIB** | Visual evaluation, LSSIM, MI, $Q^F_{AB}$, FSIM/FSIMC | Unclear |
| NSST PCNN[163] | 2018 | CT-MRI T2 | NSST, PCNN | PGIMER, **AANLIB** | STD, MI, SF, $Q^F_{AB}$, Mean | Benchmark; Unclear (cohort design) |
| NSST S-PCNN[164] | 2018 | CT-MRI-PET | HSV, NSST, ICM S-PCNN | **AANLIB** | AG, SF, MV, SD | Unclear |
| DRT[165] | 2018 | MRI-PET | DRT, SSF | **AANLIB** | MI, SF, ASSIM, $Q^F_{AB}$ | Benchmark |

Continued on next page

Table 3 continued

| Paper | Year | Modalities | Method | Datasets | Metrics | Validation evidence |
|---|---|---|---|---|---|---|
| S-ICP [166] | 2019 | 3D coronary artery-vein anatomy SPECT | S-ICP, CRT | Private | Registration error | Unclear |
| BWT [167] | 2019 | CT-MRI, MRI T1-MRA | BWT, GFS rules | **Imagefusion.org** | FF, FS, Entropy, $Q_{AB}^{F}$ | Unclear |
| CS-MCA[168] | 2019 | CT-MRI T2, MRI T1-T2 | CSR, MCA | **AANLIB** | LMI, $Q_{AB}^{F}$, $Q_{HVS}$, $Q_{E}$ | Unclear |
| CNN-DCSCM[169] | 2019 | CT-MRI | VGG, DCSCM | **AANLIB** | MSSIM, IE, MI, SF, $Q_{AB}^{F}$ | Benchmark |
| Siamese CNN[170] | 2020 | MRI-CT, MRI T1-T2, MRI-PET/SPECT | CNN, contrast pyramid | **AANLIB** | TE, MI, VIF, $Q_{AB}^{F}$, visual evaluation | Unclear |
| PMGI [171] | 2020 | MRI-PET | PGMI | **AANLIB** | CC, EN, FMI, MI, SF, SSIM, SD | Benchmark |
| MIEE-PCNN[172] | 2021 | CT-MRI, PET-MRI | MIEE, PCNN, PSO | **AANLIB**, **Imagefusion.org** | Visual evaluation, NMI, SSIM, $Q^{AB/F}$, FSIM | Unclear |
| BMGAN [56] | 2021 | MRI-PET | 3D Dense-UNet, GAN | **ADNI**, OASIS-3, **TCIA** | PSNR, FID, MS-SSIM, MAE | Unclear |
| AISA-NSST [173] | 2021 | CT-PET, MRI-CT, MR T1-T2 | NSST, AISA, MG | **AANLIB** | PSNR, SSIM, SD, EN, MI, $Q^{AB/F}$, time, visual evaluation | Benchmark |
| ECLAHE [174] | 2021 | CT-MRI | SNNE, ECLAHE | **AANLIB**, **BraTS 2015** | MSE, PSNR, SSIM, EN, F1, accuracy, recall | Benchmark |
| CS-PCA[175] | 2021 | MRI DW-FLAIR, DW-T1, FLAIR-T1, DW-FLAIR-T1 | CS, PCA, Tsallis WSS | ISLES 2015 | DSC, JI, PR, AC, SE | Benchmark |
| DCT [53] | 2021 | CT-MRI | DWT, DCT | Private | MSE, AG, PSNR, EN, SD, SNR | Unclear |
| Lambert[176] | 2022 | MRI-Fundus | 2D-3D Lambert | Private | NCC, VR, AR, distance, visual evaluation | Unclear |
| Variational optimization[177] | 2022 | CT-MRI | Variational optimization fusion | Harvard Whole Brain Atlas | MI, $Q_{W}$, $Q^{G}$ | Benchmark |
| IST-NSST [178] | 2022 | PET-CT | WAGLE, IST | **AANLIB** | VIF, EDGE, FD, AG, SF, SD | Unclear |
| CDF 9/7 NSST[179] | 2022 | CT-MRI, MRI-PET, MRI-SPECT | CDF 9/7 NSST, MPCNN | **AANLIB** | VIF, SS, EN, SD | Unclear |
| SIFI [103] | 2022 | CT-MRI, MRI T1-T2, MRI-Gad-T2 | SIFS, fuzzy set | **AANLIB** | AG, MI, Entropy, SD, Mean, visual evaluation | Unclear |
| CBF-DTF-RGF[58] | 2022 | CT-MRI | DTF, RGF, CBF | MATLAB Central File Exchange | CC, EN, SD, MI, AG, API | Benchmark |
| SwinFusion [34] | 2022 | PET-MRI, CT-MRI | Swin Transformer, CNN | **AANLIB** | PSNR, SSIM, $Q_{AB}^{F}$, FMI | Benchmark |
| PAULPCNN [180] | 2023 | PET-MRI, SPECT-MRI | PAULPCNN, NSST, DWRE | **AANLIB** | UQI, Edge, AG, MIF, SSIM, $Q_{AB}^{F}$ | Unclear |
| M3AE [12] | 2023 | MRI T1-T1c-T2-FLAIR | $M^3$AE, model inversion, self-distillation | **BraTS 2018, 2020** | DSC, HD95 | Benchmark |
| 3D U-Net [55] | 2024 | MRI T1-T1ce-T2-FLAIR | 3D U-Net, attention | **BraTS 2018** | DSC, IoU | Unclear |
| MHFN input baseline [107] | 2024 | MRI T1W–T2W-STIR | Raw-input concatenation; baseline in the MHFN study | Private | DSC, IoU, HD | Unclear (single-center cohort) |

**Abbreviations**: EN (Entropy), SD (Standard Deviation), SF (Spatial Frequency), $Q_{AB}^{F}$ (Edge Information Transfer Metric); $Q_{HVS}$ (Human Visual System-Based Metric), $Q_{E}$ (Edge Preservation Metric), $Q_{W}$ (Weighted Quality Metric), $Q_{G}$ (Gradient-Based Metric); PSNR (Peak Signal-to-Noise Ratio), MSE (Mean Squared Error), MI (Mutual Information), NMI (Normalized Mutual Information), LMI (Local Mutual Information); SSIM (Structural Similarity Index), MSSIM/MS-SSIM (Multi-Scale SSIM), LSSIM (Local SSIM), FSIM/FSIMC (Feature Similarity Index/with Color); DSC (Dice Similarity Coefficient), HD95/HD (95th Percentile Hausdorff Distance/Hausdorff Distance), IoU (Intersection over Union); F1 (F1-Score), JI (Jaccard Index), UQI (Universal Quality Index), VIF (Visual Information Fidelity), SNR (Signal-to-Noise Ratio); TMQI (Total Morphological Quality Index), FMI (Fusion Mutual Information), MIF (Modified Mutual Information), CC (Correlation Coefficient); TRE (Target Registration Error), FRE (Fiducial Registration Error), MAE (Mean Absolute Error), RMSE (Root Mean Square Error), NCC (Normalized Cross-Correlation).

Table 4: Representative feature-level fusion studies. Validation evidence is defined in Section 6.4.

| Paper | Year | Modalities | Method | Datasets | Metrics | Validation evidence |
|---|---|---|---|---|---|---|
| Shunt color[181] | 2001 | MRI-PD, MRI-SPECT | Shunt Neural Network, Color Opponent Fusion | **AANLIB** | Visual evaluation | Unclear |
| Wavelet transform [182] | 2001 | PET-MRI, CT-MRI | Wavelet Pyramid, Mallat Orthogonal Wavelet Decomposition, Max Selection | Private | Bias, visual evaluation | Unclear |
| 3D convolution[183] | 2002 | MRI PD-T1-T2, MRI-SPECT | Shunt Neural Network, Color Opponent Fusion | **AANLIB** | Visual evaluation | Unclear |
| Color table fusion [149] | 2004 | MRI-CT-PET | 2D CLT, blending | Private | Visual evaluation | Unclear |
| PCNN [184] | 2005 | CT-MRI, CT-SPECT, MRI-PET | DWF, PCNN, OMP | **AANLIB** | MI, EN, Visual evaluation | Unclear |
| 2d DWT[59] | 2006 | CT-MRI | DWT, IDWT | Private | RMSE, PSNR | Unclear |
| Regis+ICP[150] | 2007 | DE-MRI&PV-MRA, PV-MRA&EAM, DE-MRI&PV-MRA&EAM | ICP | Private | Registration error | Unclear |
| Wavelet transform[185] | 2010 | CT-MRI, MRI-PET | DWT, Max Selection | Private | Mean, Variance, EN, CEN, visual evaluation | Unclear |
| FDCT [186] | 2012 | CT-MRI | DWT, FDCT, IDWT, IHS Transform | Private | RMSE, PSNR | Unclear |
| NSCT [187] | 2013 | CT-MRI | NSCT, MSML | Private | MI, $Q^{AB/F}$, SF | Unclear |
| DWT-PCA [152] | 2013 | CT-PET | PCA, DWT | Private | MSE, PSNR, EM, visual evaluation | Unclear |
| Framelet HVS[188] | 2013 | CT-MRI, MRI-PET | Framelet Transform, HVS, IHS Transform | **AANLIB** | Mean, EN, SF, MI, SSIM, $Q^{AB/F}$ | Unclear |
| DWT [189] | 2015 | PET-MRI | DWT, FCM, IHS Transform | **AANLIB** | MSE, PSNR, SD, AG | Unclear |
| MST-SR[61] | 2015 | CT-MRI, Gd DTPA MRI-MRI T2 | MST, Sparse Representation, OMP | Private | SD, EN, QG, QP, QW | Unclear |
| Surface regis[190] | 2015 | CT-US | EM System, ICP Algorithm | Private | RMS | Unclear |
| QC hybrid [156] | 2015 | CT-MRI | DWT, PCA | Private | LM, E, MI, visual evaluation | Unclear |
| SOMP [191] | 2016 | MRI-CT, MRI-MRA, MRI-PET | K-SVD, SOMP, Sparse Representation | **AANLIB**, **Imagefusion.org** | EN, EI, $Q^{AB/F}$, MI, VIF | Unclear |
| Wavelet channel fusion [160] | 2017 | MRI T1-CT, MRI DWI-CT | DWT, Mallat Algorithm, Max Selection | Human Connectome Project | IE, MI, MG, SF, visual evaluation | Unclear |
| PCA-DWT [161] | 2017 | CT-MRI, MRI-SPECT | PCA, DWT, PCNN | **AANLIB** | STD, E, PSNR, MI, $Q^{AB/F}$ | Unclear |
| NSCT-PCNN[192] | 2017 | CT-PET | NSCT, PCNN, NSML | CT | AG, EN, IQI, $Q^{AB/F}$ | Unclear |
| Pyramid CNN[65] | 2017 | CT-MRI | Siamese CNN, image pyramids | SPECT | EN, FMI, $Q_G$, $Q_E$, VIFF | Unclear |
| MFDF-NSST[193] | 2018 | CT-MRI | NSST, MFDF, SML, Max Selection | **AANLIB**, **Imagefusion.org** | MI, EN, $Q_E$ | Unclear |
| CNN Type-II (feature)[32] | 2018 | PET-CT-MRI T1/T2 | Separate 2D CNN encoders; shared classifier | **TCIA** | Pixel-wise Accuracy, visual evaluation | Benchmark |
| MANOVA [194] | 2018 | fMRI-sMRI-DTI | Multivariate analysis of variance | Private | Jaccard, Kappa, Wilk's lambda, ROC, AUC | Unclear (single-center cohort) |
| DCHWT-PCNN [195] | 2018 | MRI-PET, MRI-SPECT | DCHWT, PCNN | **AANLIB** | AG, STD, SSIM, $Q^{AB/F}$, MI, FSIM | Unclear |
| SKR-LDP [64] | 2018 | PET-MRI, SPECT-MRI | Sparse Representation, OMP, PCA, K-SVD | **AANLIB** | MI, $Q^{AB/F}$, VIF, $Q_{CB}$ | Benchmark |
| Fuzzy-RBFNN [196] | 2018 | CT-MRI, MRI-SPECT | Fuzzy-RBFNN, EBPA-GSA, PCNN | Private | EN, STD, MI, $Q^{AB/F}$, MSE | Unclear |
| Tetrolet SUnSAL [63] | 2018 | CT-MRI, PET-MRI, SPECT-MRI | SUnSAL, Tetrolet Transform, Cycle-spinning | **AANLIB**, **Imagefusion.org** | MI, LSSIM, $Q^{AB/F}$, FSIM, FSIMC | Unclear |
| MDWT [197] | 2018 | CT-PET, SPECT, MRI, MRA, MRT | MDWT, PCA, Max Selection | Private | CE, H, SD, SF, FQI, RO, AVGER | Unclear |
| FCN [198] | 2019 | PET-CT | FCN, V-Net | Private | DSC, CE, VE | Unclear |
| NSCT [199] | 2019 | CT-MRI, MRI-PET, SPECT-MRI | NSCT, YUV Transform | **AANLIB** | TE, IE, $Q^{AB/F}$, $Q_P$, MI, $Q_Y$, $Q_{CB}$, VIF | Unclear |
| MCFO [200] | 2019 | MRI-PET, CT-PET, MRI-CT | DWT, MCFO, PCA | MedPix Dataset | AVG, L, C, STD, E, PSNR, $Q^{AB/F}$ | Unclear |

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Table 4 continued

| Paper | Year | Modalities | Method | Datasets | Metrics | Validation evidence |
|---|---|---|---|---|---|---|
| NSST-SPCNN[201] | 2020 | CT-MRI, CT, MRI | NSST, SPCNN , RE, EOG | Private | SD, EN, MI, SF, $Q_G$, $Q_{CB}$ | Unclear |
| FCM-PCA CNN[202] | 2019 | CT-MRI, PET-MRI, SPECT-MRI | CNN, FCM, PCA | AANLIIB | EN, MI, STD, UIQI, FMI, SF | Unclear |
| Multi-level fusion [203] | 2020 | PET-CT | Image-, matrix-and feature-levels fusion | **TCIA** | C-index, Spearman, log-rank p, Kaplan-Meier | Unclear (multicenter design) |
| IVD-Net [204] | 2019 | MRI multi-sequence | U-Net, HyperDenseNet, Inception | MICCAI 2018 challenge | DSC | Unclear |
| Patch CNN[205] | 2019 | PET-CT-MRI T1, PET-CT-MRI T2 | multi-stage CNN fusion | STS | DSC, Acc | Unclear |
| ZL CNN[206] | 2019 | MRI-CT/SPECT/PET, MRI T1-MRI T2-CT-PET | CNN, VGG-16, NSCT, PCNN | **AANLIB** | EN, MI, SSIM, $Q_{MRI-PET}$ | Unclear |
| SWT-FL[207] | 2019 | CT-MRI, MRI T1-T2, MRI T2-SPET | SWT, FLICM | Private | MI, $Q^{AB/F}$, BSSIM, SSIM | Unclear |
| Siamese CNN [170] | 2020 | MR-CT, MR-T, MR-PET, MR-SPECT | CNN, contrast pyramid | **AANLIB** | $Q^{AB/F}$, MI, VIF | Unclear |
| U2Fusion[208] | 2022 | PET-MRI | Unsupervised loss, VGG, EWC | **AANLIB** | CC, SSIM, PSNR, SCD, EI, VIF, MG, visual evaluation | Unclear |
| NSML-NSST[209] | 2020 | CT-MRI, MRI T1-T2, MRI T2-GAD, MRI-PET, MRI-SPECT | NSST, YIQ Transform, Mamdani | **Imagefusion.org**, **AANLIB** | EN, FMI, $Q^{AB/F}$, SD, MSSIM | Benchmark |
| MM ASeDiL[210] | 2020 | CT-MRI, MRI T1-MRA, MRI Gd-T2, MRI T1-T2 | NSCT, ASeDiL, Gabor Filter | **AANLIB** | Piella–Heijmans, SF, UIQI, MI | Unclear |
| DWT0-BCSO[211] | 2020 | MRI-CT | DWT, BCSO | **AANLIB** | EN, SSIM, FF | Benchmark |
| Auto CNN[113] | 2020 | MRI-CT | SIFT, DWT, Curvelet Transform, interpolation, CNN | Private | Acc, visual evaluation | Unclear |
| NMHE-SSGSM[62] | 2020 | CT-MRI | CSR, Average Rule, Max Selection | **Imagefusion.org**, **AANLIB** | MI, $Q^{AB/F}$, FMI, EN, VIF | Unclear |
| IFCNN[212] | 2020 | CT-MRI | IFCNN, CNN, ResNet101 | Private | VIFF, ISSIM, NMI, SF, AG | Unclear |
| LRD [213] | 2020 | PET-MRI, SPECT-MRI | LRD, OD, NOD, IRS, LEM, LDM | **AANLIB** | STD, MI, UQI, TMQI | Unclear |
| 3D MM-NAS [94] | 2020 | PET-CT | MM-NAS, CNN | **TCIA** | Acc, Prec, F1, AUC, ROC | Benchmark |
| NSST+BM-PCNN [214] | 2025 | MRI-PET, SPECT | NSST, BM-PCNN, MSMG | **AANLIB** | EN, SD, NMI, VIF, $Q_F$ | Unclear |
| MMFNet[215] | 2020 | MRI T1-T2-T1ce | MMFNet, 3D-CBAM, RFBlock | Private | DSC, ASD, HD | Unclear (single-center cohort) |
| Res-DenseNet [216] | 2020 | MRI T1-T1ce-T2-FLAIR | DenseNet, ResNet | **BraTS2015, BraTS2018** | DSC | Benchmark |
| MGMDcGAN[101] | 2020 | MRI-PET, MRI-SPECT, CT-SPECT | cGAN, mask | **AANLIB** | EN, SF, MG, PSNR, SSIM | Unclear |
| Bilevel fusion[82] | 2021 | CT-MRI, MRI-SPECT | CNN, Transformer bilevel optimization | **AANLIB** | VIF, AG, EN, SF, SCD, $Q^{AB/F}$ | Unclear |
| DDcGAN[81] | 2020 | MRI-PET | DDcGAN, dual-discriminator | **AANLIB** | EN, MG, SF, SD, PSNR, CC, SSIM, VIF | Benchmark |
| DBED [217] | 2021 | CT-MRT, MRGAD-MRT, CT-MRPD | GAN, DBED | **AANLIB** | MI, SSIM, NCIE, Chen-Blum, visual evaluation | Benchmark |
| NSCT-Xception[218] | 2021 | MRI, CT | NSCT, Xception image reconstruction | **Imagefusion.org**, **AANLIB** | API, EN, AG, MI, FS, CC, SF, RMSE, PSNR | Benchmark |
| EMFusion[2] | 2021 | CT, MRI, PET, SPECT | FusionNet, CNN | **AANLIB** | SSIM, PSNR, CC, VIF | Benchmark |
| EOA Gabor energy[219] | 2021 | MRI-PET | DSWT, FR-MGE, EOA | **AANLIB** | ALI, CI, S, $Q^{AB/F}$, VIFF, FMI | Unclear |
| BMGAN[56] | 2021 | MRI-PET | 3D Dense-UNet, GAN | ADNI, OASIS-3, **TCIA** | MAE, PSNR, MS-SSIM, FID | Unclear |
| 3D CNN[220] | 2021 | MRI-fMRI-DTI | 3D-CNN | COBRE | ACC, SEN, SPE, AUC | Unclear |
| DTNP-NSCT[221] | 2021 | CT-MRI, MRI T1-T2, MRI-PET/SPECT | NSCT, INSML, WLE, DTNP | **AANLIB** | Petrovic, MI, FMI, SCD, MS-SSIM, SD | Unclear |

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Table 4 continued

| Paper | Year | Modalities | Method | Datasets | Metrics | Validation evidence |
|---|---|---|---|---|---|---|
| NSCT-CNN[222] | 2021 | CT-MRI, MRI multi-sequence | NSCT, CNN, PHF-CNN, FCN-8s | **AANLIB** | MI, SSIM, VIF, $Q^{AB/F}$ | Unclear |
| CBF-ANFIS [218] | 2021 | CT-MRI, MRI GAD-T2, CT-MR GAD, MR GAD-PET-FDG | CBF, ANFIS, DWT, Average Rule | **AANLIB** | API, AG, EN, MIF, FS, CC, SF, PSNR, $Q^{AB/F}$ | Benchmark |
| NSST-Co-LE[60] | 2021 | CT-MRI, MRI-PET | NSST, LE, CoF, MSS, SML, Max Selection | **AANLIB** | MI, SD, $Q^{AB/F}$, SF, EN | Benchmark |
| DCT [53] | 2021 | CT-MRI | DCT, DWT | Private | SNR, PSNR, MSE, AG, SD, EN | Unclear |
| Aggregated NAS[95] | 2021 | MRI-PET | NAS, layered fusion | **AANLIB** | SD, EN, VIF, CC, SCD | Unclear |
| DPCN [223] | 2021 | SPECT-MRI | CNN, SFFEB, FSFEB | **AANLIB** | MI, VIF, FMI, $Q_P$, $Q_Y$, time | Unclear |
| MSRPAN [104] | 2021 | MRI-CT, MRI-PET, SPECT | CNN, MSRPAN, FER | **AANLIB** | $Q_{AC}$, SSIM, MI, PSNR, $Q_{AC}$, TMQI, FSIM, time | Unclear |
| SDNet[224] | 2021 | PET-MRI | SDNet, adaptive decision | **AANLIB** | EN, $Q^{AB/F}$, FMI, PSNR, MG, time, visual evaluation | Unclear |
| PPT fusion[225] | 2021 | MRI-PET | Patch/Pyramid Transformer, MLP | **AANLIB** | EN, SSIM, MS-SSIM, MI, FMI, SCD, CC, SD, VIF, visual evaluation | Unclear |
| IGNFusion [226] | 2022 | CT-MR T1, PET-MRI T2 | VGG-16, MSCAFM/MSCAFM, IGM, SSW | **AANLIB** | NMI, $Q^{AB/F}$, $Q_Y$, UQI | Unclear |
| DB-NSCT [227] | 2022 | CT-MRI, MRI-PET | NSCT, Davies-Bouldin | **AANLIB** | MI, SSIM, $Q^{AB/F}$ | Unclear |
| FDO [228] | 2023 | MRI-PET | SVD, WMC, FDO, DPGF | **AANLIB** | AVG, MI, $Q^{AB/F}$, VIF, SSIM, PSNR | Unclear |
| COVID-19 CNN[229] | 2022 | CT-X ray | AlexNet, Vgg16, GoogLeNet | Private | ACC, SEN, SPEC, MCC, AUC | Unclear |
| CRFNet [111] | 2024 | Acetic-acid–iodine colposcopy | Proposal shifting; corresponding channel attention | Private | Average precision | Unclear |
| ST-cGAN [57] | 2022 | MRI T1-T2 | cGAN | IXI | PSNR, SSIM, FSIM, MMD | Unclear |
| MUM-Net [230] | 2022 | MRI-US | MUM-Net, Disentangling modality features | Private | AUC, ACC, SENS, SPEC, PREC, Grad-CAM | Unclear |
| ADSR [231] | 2022 | MRI T2+FDG-PET, MRI T2+SPECT-TI/TC | Sparse Representation, PCA, K-SVD | **AANLIB** | EN, AG, MI, $Q_W$, $Q^{AB/F}$ | Unclear |
| IoMT-NSST [232] | 2022 | CT-MRI, MRI-SPECT | NSST, MLE | **AANLIB** | MI, SD, $Q^{AB/F}$, SF, Mean, AG, CC, time | Benchmark |
| MATR[117] | 2022 | CT-MRI, MRI-SPECT/PET | Multiscale adaptive Transformer, composed loss | **AANLIB** | MI, TE, NICE, G, P, CV, LMI, VIF, MS-SSIM | Unclear |
| DualMMP-GAN[46] | 2022 | MRI T1-T2/FLAIR | DualMMP-GAN, GMMS | **BraTS 2018** | PSNR, SSIM, RMSE, DSC, Sensitivity, Spesificity, HD95 | Benchmark |
| MMRACR-Net [233] | 2022 | MRI DWI-ADC | MMRAM, CCM | Private | Acc, AUC, SEN, SPC | Unclear |
| TAUNet[234] | 2022 | MRI bSSFP-LGE | CNN, Transformer, OCR | MyoPS 2020 | DSC, HD95 | Benchmark |
| SwinFusion[34] | 2022 | PET-MRI, CT-MRI | CNN, Swin Transformer, ACFM | **AANLIB** | FMI, $Q^{AB/F}$, SSIM, PSNR | Benchmark |
| DILRAN[235] | 2022 | CT-MRI | CNN, DILRAN, Softmax | **AANLIB** | PSNR, SSIM, FSIM, MI, FMI, EN | Benchmark |
| IFT [236] | 2022 | MRI-PET | Spatio-Transformer, RFN-Nest | **AANLIB** | EN, SCD, MI, MS-SSIM, SD, CC, MG | Unclear |
| MRSCFusion[118] | 2023 | CT-MRI, PET-MRI, SPECT-MRI | Swin Transformer, CNN, RSCF, AWB | **AANLIB** | SCD, $Q^{AB/F}$, FMI, SF, VIF | Unclear |
| DFENet[237] | 2023 | CT-MRI, PET-MRI, SPECT-MRI | CNN, Transformer, DFENet, GSFAM, LEGM | **AANLIB** | MI, FMI, NICE, SSIM, $Q^{AB/F}$, CV, $Q_{CB}$ | Benchmark |
| Hybrid Transformer[238] | 2023 | PET-MR, SPECT-MR | CNN, Transformer, self-adaptive weight fusion rule | **AANLIB** | MI, $Q_P$, $Q_{NICE}$, SSIM | Benchmark |
| F²Net[239] | 2023 | MRI multi-sequence | Transformer, PVTv2, F²Net, CFM, MCM | **BraTS 2019, BraTS 2020**, ISLES 2015 SISS | DSC, IoU, HD95, Sen | Unclear |
| Epsilon GFS [240] | 2023 | CT-MRI | Shearlet Transform, Guided Filter, Max Selection, Average Rule | Public MRI-CT dataset | $Q^{AB/F}$, $L^F_{AB}$, $N^F_{AB}$, MAD, SD | Unclear |

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Table 4 continued

| Paper | Year | Modalities | Method | Datasets | Metrics | Validation evidence |
|---|---|---|---|---|---|---|
| U-Patch GAN[86] | 2023 | SPECT TI-MRI T2, SPECT TC-MRI T2, CT-MRI T2, PET FDG-MRI T1 | PatchGAN, spectral norm | **AANLIB** | $Q_G$, $Q_Q$, FMI, SSIM, RMSE, MI | Unclear |
| ELF[241] | 2023 | 2D Fundus-3D OCT | CNN, ELF | GAMMA | Acc, Cohen's Kappa | Benchmark |
| AMTNet [242] | 2023 | MRI T2-ADC, MRI T1-FLAIR | Transformer, CNN, AMTNet, R-MSA, AITF, 3D-CDS | **BraTS 2021** | DSC, HD95, Jaccard, RVD | Benchmark |
| DDFM[71] | 2023 | MRI-CT, MRI-PET, SPECT-MRI | DDPM, hierarchical Bayesian inference | **AANLIB** | EN, SD, MI, VIF, $Q^{AB/F}$, SSIM | Benchmark |
| MSU-Net[243] | 2023 | MRI T1-T2-T1ce | U-Net, MSU-Net, SCIEM, MSFEM | **AANLIB**, private | DSC, PREC, Recall, IOU | Unclear |
| CDDFuse[9] | 2023 | MRI-CT, MRI-PET, MRI-SPECT | CNN, Restormer, INN, CDDFuse | **AANLIB** | EN, SD, SF, MI, SCD, VIF, $Q^{AB/F}$, SSIM | Benchmark |
| IE-CFRN[244] | 2023 | CT-MRI T1/T2, MRI T1-GAD, MRI T1-PD, MRI-PET/SPECT TC /SPECT TI | CNN, GCIE, HFEN, LWEN | **AANLIB** | $Q^{AB/F}$, VIFP, IFC, NMI | Unclear |
| CAE+MCIEN[10] | 2023 | CT-MRI T1/T2, MRI T1-T2, MRI T2-GAD, MRI T2-PD, MRI T2-PET/SPECT TC/SPECT TI | CNN, Vision Transformer, CAE, MCIEN | **AANLIB** | NMI, MI, FSIM, PC, GM, $Q^{AB/F}$, UIQI | Unclear |
| 3D THERMO-SCAN[245] | 2023 | CT-Infrared | CNN, 3D THERMO-SCAN | Private | Registration error, visual evaluation | Unclear |
| TUFusion[39] | 2024 | Medical and other image domains; exact medical pairing unverified | CNN–Transformer encoder; composite attention | Medical subset not verified | PSNR, RMSE, SSIM | Unclear |
| SSTFusion [246] | 2023 | CT-MRI | LeWin Transformer, CNN, RSR pretext | **AANLIB** | MI, PSNR, $Q^{AB/F}$, SSIM, AG, EI, SF | Unclear |
| ALMF-Net[96] | 2024 | MRI-PET/CT/SPECT | NAS, DARTS, mask | **AANLIB**, **BraTS 2021** | MI, CC, $Q^{AB/F}$, VIF, P, FLOP, L | Unclear |
| MUFusion[36] | 2023 | PET-MRI | VGG-19, memory unit | **AANLIB** | VIF, EN, MI, EI, $Q^{AB/F}$, MSSIM, DSC | Unclear |
| M3AE [12] | 2023 | MRI T1-T2-T1ce-FLAIR | M³AE, self distillation, model inversion | **BraTS 2018/2020** | DSC, HD95 | Benchmark |
| FDGNet[247] | 2023 | CT-MRI, PET-MRI, SPECT-MRI | CNN, FDGNet | **AANLIB** | NMI, QY, $Q^{AB/F}$, VIF | Benchmark |
| MURF[114] | 2023 | PET-MRI, CT-MRI | Mutually reinforcing registration-fusion, Shared Information Extraction via contrastive learning, Multi-scale Coarse Registration | **AANLIB** | RMSE, MAE, MEE, NCC, MG, EI, VIF, SSIM, MI | Unclear |
| M4FNet[98] | 2023 | CT-MRI, PET-MRI, SPECT-MRI | CNN, M4FNet | **AANLIB**, **BraTS 2019** | EI, SD, $Q^{AB/F}$, MI, MSSIM, PC, VIF, UIQI | Unclear |
| FATFusion[67] | 2024 | SPECT-MRI, PET-MRI | Transformer, FGTM, AGTM | **AANLIB** | NMI, QY, $Q^{AB/F}$, VIF, Chen-Blum | Unclear |
| MCF-Net[91] | 2024 | MRI multi-sequence | ViT, mask align, sobel | **BraTS 2020** | PSNR, SSIM, NMSE | Unclear |
| MACTFusion[105] | 2025 | CT-MRI, PET-MRI, SPECT-MRI | ViT, CNN multi-axis attention | **AANLIB** | SCD, SD, MS-SSIM, $Q^{AB/F}$, VIF, $Q_w$ | Unclear |
| F-DARTS[248] | 2023 | CT-MRI, MRI-PET, MRI-SPECT, multi-sequence MRI | Unsupervised differentiable architecture search, HVS-aligned kernel learning, unsupervised loss, channel attention in search space | **AANLIB**, BrainWeb, RIRE | MI, $FMI_w$, $Q^{AB/F}$, SCD, VIF, SD, $Q_{CB}$ | Unclear |
| ALYOLOv5 [7] | 2024 | PET-CT | YOLOv5, CFF , AAF | Private | Precision, recall, mAP, F1 | Unclear (single-center cohort) |

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Table 4 continued

| Paper | Year | Modalities | Method | Datasets | Metrics | Validation evidence |
|---|---|---|---|---|---|---|
| MGCNet[249] | 2024 | Multi-sequence MRI | Guided local feature fusion; cross-modal complementary external attention | Private | ACC, AUC, C-index, HR | Unclear |
| EMMA [250] | 2024 | MRI-CT, MRI-PET, SPECT-MRI | U-Net, Restormer, EMMA | **AANLIB** | EN, SD, SF, AG, SCD, VIF | Benchmark |
| SRMNet[251] | 2024 | MRI multi-sequence | Missing-modality segmentation, Local-global feature enhancement, Deformation-Adaptive Perception fusion | **BraTS 2018**, **BraTS 2020** | DSC, HD95 | Benchmark |
| CoCoNet[140] | 2024 | MRI-PET, MRI-SPECT | VGG, contrastive learning, CoCoNet | **AANLIB** | EN, AG, SF, SD, SCD, VIF | Benchmark |
| ASFE-Fusion [66] | 2024 | CT-MRI, PET-MRI, SPECT-MRI | CNN, CASF, AFF, SFEF, FFT | **AANLIB** | EN, PSNR, MI, CC, FMI, Prec, Recall, F1, Acc | Unclear |
| CA-DDPM [73] | 2024 | MRI-CT | DDPM, CNN, Transformer | **AANLIB** | EN, MI, SCD, SSIM, VIF, $Q^{AB/F}$ | Benchmark |
| Diff-IF[90] | 2024 | MRI, CT, PET, SPECT | DDPM, prior knowledge, target search | **AANLIB** | MI, SCD, VIF, $Q^{AB/F}$, SSIM | Unclear |
| TIEF [252] | 2024 | MRI multi-sequence, SPECT-MRI, CT-MRI | CNN, SDCL, TIEF | **BraTS 2019**, **AANLIB** | $Q^{AB/F}$, EN, AG, SF, CI, MI, PSNR, SSIM, F | Unclear |
| Mambadfuse[70] | 2024 | MRI-PET, MRI-SPECT | Mamba, CNN | **AANLIB** | EN, SD, SF, MI, SCD, VIF, $Q^{AB/F}$, SSIM, mAP, Grand-CAM, visual evaluation | Benchmark |
| Fuse4Seg[108] | 2024 | MRI T1-FLAIR-T2-T1ce, MRI-CT/PET/SPECT | SwinUNet, dual-level optimization | **BraTS 2021**, **AANLIB** | EN, SD, SF, MI, SCD, VIF, $Q^{AB/F}$, DSC, mIoU, HD95, visual evaluation, attention heatmap | Benchmark |
| MHFN [107] | 2024 | MRI T1W–T2W-STIR | Hierarchical feature fusion with AGFM | Private | DSC, IoU, HD | Unclear (single-center cohort) |
| CDE-Net[84] | 2024 | MRI T1-T2-FLAIR-T1ce | CDE-Net, DST, CDFL, RW-PAM | **BraTS 2021** | DSC, HD95, visual evaluation | Unclear |
| TIMFusion[253] | 2024 | MRI-CT/PET/SPECT | Task guidance constraints, IAS, PMI | **AANLIB** | MI, FMI, VIF, $Q^{AB/F}$, EN, SCD, parameter, time | Benchmark |
| LRFNet[37] | 2024 | MRI-PET, MRI-SPECT | LRU, FEF, DGSSIM | **AANLIB** | EN, SD, MI, AG, CC, SCD, $Q^{AB/F}$, SF | Unclear |
| CMINet[254] | 2025 | CT-MRI, MRI-PET, MRI-SPECT | Transformer, CNN, GDA/GADN | **AANLIB** | MI, SF, VIF, $Q^{AB/F}$ | Unclear |
| PDMINet[255] | 2025 | MRI multi-sequence | Prototype-driven + multiexpert integration | **BraTS 2018**, **BraTS 2019**, **BraTS 2020** | DSC, HD95, Sensitivity, Specificity | Unclear |
| GeSeNet[256] | 2024 | CT-MRI, MRI-PET, MRI-SPECT | Semantic-guided fusion, edge enhancement gradient filter, region mask for redundant computation pruning | **AANLIB** | SSIM, SD, MI, VIF, SCD, $Q^{AB/F}$ | Unclear |
| DRMF[89] | 2024 | CT-MRI, MRI-PET, SPECT-MRI | Degradation-robust conditional diffusion, fusion + regularization + mask-guided loss | **AANLIB** | MI, SF, SD, VIF, SSIM, PSNR, MUSIQ, LIQE, mAP | Unclear |
| TFEFusion[257] | 2025 | PET, MRI | CNN, Swin Transformer, TFEFusion | **AANLIB** | G, SSIM, $Q^{AB/F}$, PSNR, SF, RMSE | Benchmark |
| LAGN Diffusion[124] | 2025 | MRI-CT, MRI-PET, MRI-SPECT | Diffusion, Transformer, CNN | **AANLIB** | EN, SD, SCD, VIF, PSNR, SSIM | Benchmark; Reader study (student ratings) |
| PFRFusion [5] | 2025 | CT-MRI, MRI T1-T2 | PFAR, DBFF, co-training | **AANLIB**, Synth-Strip | MAE, NMI, MEE, MI, VIF, SSIM, IMF, SF, DSC, IoU, MPA | Unclear |
| EFCNet [92] | 2025 | MRI-SPECT, MRI-PET | EFCNet, SEEM, MSAM, E-Fusion, IRM | **AANLIB** | AG, SNR, MI, $Q^{AB/F}$, VIF, PSNR, EI, SF | Unclear |
| DTMFusion [97] | 2025 | CT-MRI, MRI T1-T2 | Transformer, Mamba, DTSDR, MCF | **AANLIB**, Synth-Strip | PSNR, NCC, MSE, SD, MI, VIF, SCD | Unclear |
| EDPSN [125] | 2025 | MRI-CT, MRI-PET, SPECT-MRI | EDPSN, EEM, ED, DPS, VGG16 | **AANLIB**, CHAOS | EN, SD, SCD, $Q^{AB/F}$, VIF, PSNR, SSIM | Benchmark; Unclear (reader protocol) |

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Table 4 continued

| Paper | Year | Modalities | Method | Datasets | Metrics | Validation evidence |
|---|---|---|---|---|---|---|
| AAHN [258] | 2025 | PET-CT, MRI T1-FLAIR -DWI-ADC, CT-MRI | CNN, Transformer, GCN | Hecktor21, Prostate158, **BraTS 2019/2023/2024**, CHAOS | DSC, HD95 | Unclear |
| MLFuse[259] | 2025 | CT-MRI, PET-MRI, SPECT-MRI | CNN, Transformer, sobel | **AANLIB** | VIF, SCD, $Q^{AB/F}$, SSIM, FMI | Unclear |
| DMCMFuse[68] | 2025 | CT-MRI, MRI T1-T2, MRI-PET/SPECT | CNN, Mamba, MCFM, HAED | **AANLIB** | SSIM, VIFF, $Q^{AB/F}$, SCD, FMI, $Q_w$ | Unclear |
| FAB-Transformer[260] | 2025 | PET-MRI, SPECT-MRI | CNN, Swin Transformer, DSM, INN | **AANLIB** | MI, SF, VIF, $Q^{AB/F}$, SSIM, mAP | Benchmark |
| AdvTCUnet [87] | 2025 | 3D MRI T1-T1ce-FLAIR | Adversarial learning, Transformer, U-Net, CV energy function | **TCIA** | DSC, HD, visual evaluation | Benchmark |
| MTG-Fusion[31] | 2025 | CT-MRI, PET-MRI, SPECT-MRI | Auxiliary text guidance, image-feature fusion | **AANLIB** | EN, SD, SF, SCD, VIF, Q | Unclear |
| MAINet[261] | 2025 | MRI-CT, MRI-PET, MRI-SPECT | CMI, DGL, BUGF | **AANLIB** | MI, $Q_{NCIE}$, SD, VIF, $Q^{AB/F}$, SSIM, $Q_{CB}$, visual evaluation | Unclear |
| Uncertainty-seg[83] | 2025 | MRI FLAIR-T2 | MC-Dropout, uncertainty weighting | BraTS subset | DSC, IoU, SE, PPV, uncertainty | Unclear |
| BSAFusion[11] | 2025 | Misaligned CT-MRI, PET-MRI | MDF-FR, BSFA, MMFF | **AANLIB** | $Q^{AB/F}$, $Q_{CV}$, SSIM, VIF, $Q_S$, visual evaluation | Benchmark |
| UniFuse[38] | 2025 | MRI T1-T2/CT, CT-PET | Spatial Mamba, LoRA, DAPL, OUFR | **BraTS 2020**, SynthRAD2023, FDG-PET/CT | MSE, CC, PSNR, VIF, SSIM, GFLOP, visual evaluation | Unclear |
| ReFusion[262] | 2025 | CT-MRI, MRI-PET | Meta-learning, learnable fusion loss, inner-outer-alternate training strategy | **AANLIB** | EN, SD, SF, AG, SCD, SSIM, VIF, $Q_{CB}$, $Q_{NCIE}$ | Benchmark |
| DM-FNet[72] | 2025 | CT-MRI, MRI-PET, MRI-SPECT | Two-stage diffusion hybrid loss (intensity + SSIM + gradient) | **AANLIB** | SF, SD, AG, $Q_W$, SCD, VIFF, $Q^{AB/F}$, MSSSIM, FMI-WT | Unclear |
| C2RF[263] | 2025 | PET-MRI | Mutually promoting registration-fusion, commonality mining for feature disentanglement, multi-level registration, contrastive learning with hard negative sample mining | **AANLIB** | MI, CC, SSIM, MSE, MEE, MAE, EN, MI, SF, VIF, $Q^{AB/F}$, SSIM | Unclear |
| HCFMaNet[99] | 2026 | CT-MRI, PET-MRI, SPECT-MRI, MRI multi-sequence | Local-aware Mamba, Holistic Cross-modal Fusion Mamba, end-to-end unsupervised fusion | **AANLIB**, BrainWeb, **SynthRAD** | $Q_{NCIE}$, MI, $Q_G$, $Q_{CV}$, LMI, MS-SSIM, VIF, NCC, SSIM | Unclear |
| FlexiD-Fuse[264] | 2026 | CT-MRI, MRI-PET, MRI-SPECT, MR-T2/MR-Gad/PET, CT/MR-T2/SPECT | Diffusion model + hierarchical Bayesian modeling, Expectation-Maximization, ultra-lightweight Diffusion Fusion Mamba | **AANLIB** | SSIM, MI, PSNR, VIF, $Q_S$, $Q_{CV}$, FSIM, $Q_{NCIE}$ | Benchmark |

Abbreviations are defined below Table 3.

Table 5: Representative decision-level fusion studies. Validation evidence is defined in Section 6.4.

| Paper | Year | Modalities | Method | Datasets | Metrics | Validation evidence |
|---|---|---|---|---|---|---|
| DST-cls[77] | 1996 | Double echo MRI | DS theory, mass function, composite hypothesis | Private | K, decision consistency, visual evaluation | Unclear |
| CNN Type-III (decision)[32] | 2018 | PET-CT-MRI T1/T2 | CNN, Majority voting | **TCIA** | Pixel-wise Accuracy, visual evaluation | Benchmark |
| Patch CNN[205] | 2019 | PET-CT-MRI T1 | Voting, CNN | STS | DSC, Pixel-wise Accuracy | Unclear |
| DCMA [112] | 2020 | MRI-PET | DCNN ensemble, Adaboost, dropout | **ADNI** | Acc, Sen, Spe, AUC | Benchmark |
| Evidence fusion[35] | 2022 | MRI T1-T1Gd-T2-FLAIR | DST, contextual discounting, MMEF module | **BraTS 2021** | DSC, HD, ECE, visual evaluation | Benchmark |
| ETSCL [79] | 2024 | CFP-OCT-Vascular | Supervised contrastive loss, DST-based fusion | GAMMA | Cohen's kappa, Accuracy | Benchmark |
| MHFN decision baseline [107] | 2024 | MRI T1W–T2W-STIR | Combines modality-specific segmentation outputs; baseline in the MHFN study | Private | DSC, IoU, HD | Unclear (single-center cohort) |
| Deep evidential fusion[3] | 2025 | PET-CT, MRI multi-sequence | DST, evidence mapping, contextual discounting | PET-CT lymphoma dataset, **BraTS 2021** | DSC, Brier score, NLL, ECE | Benchmark; Retrospective |
| MAMS[122] | 2025 | FFDM-DBT | cGAN, MAMS fusion | NFH, GDHCM | RMSE, PSNR, UQI, MS-SSIM, Acc, AUC, F1-Score | Retrospective (two centers) |
| Uncertainty-seg[83] | 2025 | MRI FLAIR-T2 | MC-Dropout, uncertainty weighting | BraTS subset | DSC, IoU, SE, PPV, uncertainty | Unclear |

Abbreviations are defined below Table 3.

**Table 6**
Cross-generation comparison of representative MMIF methods under their reported configurations.

| Method | Registration assumption | Paired-data requirement | Missing modalities | Supervision | Training and inference cost | Robustness evidence and limits |
|---|---|---|---|---|---|---|
| MST-SR (2015) [61]; Transform / sparse coding | Pre-registered inputs | Training: natural-image dictionary patches; inference: paired source images | Both inputs required; no missing-input mechanism | Dictionary learning and coefficient rules; no fused targets | Offline dictionary learning; transforms and iterative patch coding at inference | Aligned medical pairs; noise sensitivity and cost depend on decomposition and patch settings |
| EMFusion (2021) [2]; CNN | Corresponding spatial content; no learned alignment module | Corresponding sources for fusion constraints and inference | Both inputs required; no missing-input mechanism | Source saliency and pretrained feature constraints; no fused targets | Pretrained encoder and network training; feed-forward inference | Structural/functional preservation and chrominance enhancement; missing-input tolerance not established |
| SwinFusion (2022) [34]; Transformer | Corresponding grids; attention does not establish registration | Paired sources in training and inference | Both inputs required; no missing-input mechanism | Source structure, texture and intensity losses; no fused targets | Network training; window attention and reconstruction at inference | Multiple fusion tasks tested; arbitrary deformation and missing-input tolerance not established |
| MambaDFuse (2024) [70]; State space model | Corresponding grids; no explicit alignment module | Paired sources in training and inference | Both inputs required; no missing-input mechanism | Source structure, texture and intensity losses; no fused targets | Network training; selective scans and convolution at inference | Medical pair benchmarks; linear scan scaling does not establish 3D clinical latency |
| DDcGAN (2020) [81]; Adversarial generation | Spatial correspondence despite unequal resolutions | Training: paired medical patches; inference: both sources | Both inputs required; no missing-input mechanism | Adversarial and source-content losses; no fused targets | Generator and two discriminators in training; generator at inference | Unequal-resolution PET-MRI tested; resolution handling does not establish deformation tolerance |
| DDFM (2023) [71]; Diffusion prior | Corresponding grids for likelihood constraints | No fusion-specific training pairs; both sources condition inference | Both sources required; completion not demonstrated | Pretrained unconditional prior and source likelihood; no fused targets | Pretrained model without fine-tuning; iterative sampling and rectification | Medical pair evaluation; prior mismatch and unsupported generated detail remain concerns |
| M3AE (2023) [12]; Missing-modality learning | Co-registered MRI volumes | Complete MRI cases for masked reconstruction; available subset at inference | Training: simulated dropout; inference: nonempty subsets | Masked reconstruction, then supervised segmentation and self-distillation | Two training stages; one segmentation model and learned substitute at inference | Missing-sequence tests on BraTS; naturally incomplete training and site shifts need validation |
| BSAFusion (2025) [11]; Joint alignment and fusion | Learned stepwise feature alignment | Training: deformed registered pairs; inference: corresponding, unaligned sources | Both modalities required; absence is not the target condition | Modality, alignment and source-fusion constraints; no fused targets | Shared encoder and alignment training; deformation prediction and fusion at inference | Simulated rigid/non-rigid mismatch tested; clinical deformation shifts require validation |

Table 7: CT–T1-Gd MRI fusion and contouring results on GLIS-RT from Safari et al. [8] (mean ± SD).

| **Method** | **Backbone / principle** | **EN** | **SSIM** | $Q^{AB/F}$ | **Contouring Dice** | **Params / FLOPs** |
|---|---|---|---|---|---|---|
| GTF | Gradient transfer | 5.25±0.42 | 0.29±0.37 | 0.60±0.06 | NR | NR / NR |
| DDCT-PCA | Transform + PCA | 5.13±0.30 | 0.33±0.16 | 0.60±0.06 | 0.91±0.07 | NR / NR |
| FPDE | Fourth-order PDE | 4.77±0.34 | 0.25±0.26 | 0.37±0.10 | NR | NR / NR |
| HMSD | Multiscale decomposition | 5.58±0.23 | 0.32±0.17 | 0.60±0.07 | 0.85±0.08 | NR / NR |
| GFDF | Traditional fusion | 2.07±0.51 | 0.66±0.33 | 0.46±0.04 | NR | NR / NR |
| IVF | Traditional fusion | 2.74±0.57 | 0.13±0.05 | 0.54±0.04 | 0.88±0.07 | NR / NR |
| MEF | Multi-exposure fusion | 2.06±0.48 | 0.81±0.03 | 0.61±0.06 | 0.89±0.09 | NR / NR |
| CNN-Fuse | CNN | 2.06±0.50 | 0.66±0.31 | 0.51±0.07 | NR | NR / NR |
| SESF-Fuse | CNN autoencoder | 2.06±0.51 | 0.58±0.25 | 0.46±0.04 | NR | NR / NR |
| SwinFusion | Swin Transformer | 2.05±0.46 | 0.67±0.16 | 0.60±0.04 | 0.89±0.07 | NR / NR |
| IFCNN | CNN | 2.11±0.48 | 0.78±0.04 | 0.60±0.05 | 0.92±0.04 | NR / NR |
| U2Fusion | CNN | 2.14±0.49 | 0.15±0.06 | 0.54±0.07 | 0.90±0.07 | NR / NR |
| DSAGAN | GAN | 2.19±0.39 | 0.11±0.05 | 0.40±0.05 | 0.72±0.14 | NR / NR |
| CU-Net | Coupled unrolling | 2.83±0.55 | 0.40±0.09 | 0.31±0.04 | NR | NR / NR |
| FusionGAN | GAN | 1.64±0.30 | 0.59±0.18 | 0.35±0.03 | NR | NR / NR |
| MedFusionGAN | Residual CNN / GAN | 5.20±0.38 | 0.62±0.22 | 0.64±0.10 | 0.96±0.02 | NR / NR |

NR: not reported in the source for the listed outcome or computational cost.

Table 8: Representative software components and reproducibility records for MMIF. Tool availability does not establish a reproduced pipeline.

| Stage | Tool | Input/output | Role in MMIF | License | Reproducibility artifacts / requirements |
|---|---|---|---|---|---|
| Registration | SimpleITK | Images and masks; aligned images and transforms. | Align CT-MRI pairs and resample onto a shared physical grid. | Apache 2.0 | Save transform direction, reference grid, interpolation, optimizer settings and registration checks. |
| Registration | ANTs | Image volumes; affine transforms and deformation fields. | Provide an alternative registration and MRI bias-correction component. | Apache 2.0 | Save command lines, masks, transform order, similarity metric and parameter files. |
| ROI and display | 3D Slicer | Volumes, labels and transforms; scenes, masks and views. | Inspect alignment, edit ROIs and review source images alongside fused outputs. | BSD-style Slicer license | Save scene, extension versions, ROI edits, display windows and overlay settings; viewing alone is not clinical validation. |
| Data preparation | MONAI | Images and metadata; transformed tensors and batches. | Compose medical-image loading, caching and synchronized spatial transforms. | Apache 2.0 | Save dataset manifest, transform order, reader options, channel order, spacing, normalization and seeds. |
| Data preparation | TorchIO | Subject images and labels; transformed volumes and patches. | Prepare corresponding 3D inputs and apply shared spatial augmentation. | Apache 2.0 | Save subject definitions, sampling policy, transform parameters and random states; verify paired geometry. |
| Algorithm development | MATLAB Image Processing Toolbox | Arrays and images; processed images and metric values. | Support image processing and prototype fusion rules supplied by the researcher. | Commercial | Save MATLAB/toolbox versions and scripts; identify any additional transform or fusion-metric code and its license. |
| Training | PyTorch | Tensors and model code; weights and predictions. | Implement the chosen fusion architecture, losses, training and inference. | BSD-3-Clause | Save code revision, environment, model configuration, checkpoint hash, seeds, device and determinism settings. |
| Training | PyTorch Lightning | PyTorch models and data modules; checkpoints and logs. | Organize training loops and distributed execution. | Apache 2.0 | Save Trainer settings, callbacks, precision, strategy, resume state and data-loader configuration. |
| Training | TensorFlow / Keras | Tensors and model definitions; saved models and predictions. | Provide an alternative framework for a specified fusion model. | Apache 2.0 | Save architecture, weights, preprocessing, dependency versions, training settings and export signature. |
| Metrics | scikit-image | Source/output arrays; PSNR, SSIM and other implemented metrics. | Compute specified image-quality measures with an explicit reference convention. | BSD-3-Clause; bundled exceptions | Save function/version, data range, window settings, channel handling, masks and per-case values. |
| Metrics | TorchMetrics | Prediction/target tensors; metric states and summaries. | Evaluate supported image or downstream-task metrics within tensor workflows. | Apache 2.0 | Save exact metric class, arguments, reduction and distributed aggregation; verify any custom fusion metric separately. |
| Tracking | TensorBoard | Event logs; scalar curves and image summaries. | Track training and inspect corresponding source/output samples. | Apache 2.0 | Archive event files with run identifiers and configurations; dashboards do not replace data manifests or scripts. |
| Export | ONNX | Supported model graphs; portable graph files. | Exchange a trained fusion model between compatible runtimes. | Apache 2.0 | Record exporter/opset, input names and shapes; check output agreement against the source framework. |
| Inference | TensorRT | Supported models; optimized inference engines. | Optimize inference on supported NVIDIA GPUs. | NVIDIA SDK terms; OSS parts Apache 2.0 | Save runtime/hardware versions, precision, shape profiles and calibration data; measure export-related output changes. |
| Inference | OpenVINO | Supported models; converted models and runtime outputs. | Execute and optimize inference on supported hardware. | Apache 2.0 | Save conversion options, device, precision and runtime version; verify outputs and measure complete-pipeline latency. |

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Table 8: Representative software components and reproducibility records for MMIF (continued).

| Stage | Tool | Input/output | Role in MMIF | License | Reproducibility artifacts / requirements |
|---|---|---|---|---|---|
| Packaging | MONAI Deploy App SDK | Application operators and models; packaged applications. | Connect configured image readers, inference and output writers. | Apache 2.0 | Save application graph, package version, series-selection rules, metadata mapping and end-to-end checks. |
| Packaging | Docker Engine | Build recipes and dependencies; container images. | Package the selected preprocessing, inference and evaluation environment. | Apache 2.0; Desktop terms separate | Archive Dockerfile, dependency locks, image digest, mounts, command and host/GPU details; containers do not fix missing data or protocols. |

**Table 9**
Commonly used datasets and image resources for MMIF. Counts refer to the stated releases and units; source pages were checked on 29 September 2026.

| Dataset | Primary objective | Modality composition | Release and data scale | Source |
|---|---|---|---|---|
| AANLIB / Whole Brain Atlas | Brain anatomy and neurological disease examples for image-fusion research. | CT, MRI, PET and SPECT; combinations vary by case. | Case-based atlas. Pair counts depend on the selected cases and slices; no fixed total of prepared fusion pairs is asserted. | Atlas |
| BraTS | Brain-tumor segmentation with expert annotations. | Co-registered T1, T1ce, T2 and FLAIR MRI. | 2020 example: 369 training cases, each with four MRI sequences. Counts for other years or tasks must be reported separately. | 2020 data<br>2020 challenge report |
| TCIA | Repository of cancer imaging collections; fusion suitability depends on the selected collection. | Collection-dependent; includes multiparametric MRI and PET/CT. | Collection examples: TCGA-GBM, 262 subjects; TCGA-LGG, 199 subjects. These are collection totals, not counts of complete paired inputs. | TCGA-GBM<br>TCGA-LGG |
| ADNI | Longitudinal imaging and biomarkers for aging and Alzheimer disease research. | Structural MRI and PET; additional MRI sequences and PET tracers depend on phase and visit. | Phase- and visit-dependent cohort. Report unique participants, imaging visits and the eligible matched-modality subset separately. | Study phases<br>Data and matching guidance |
| Imagefusion.org | Legacy image-fusion examples cited in the literature; medical subset provenance needs verification. | Study-specific image pairs; a current medical release could not be verified. | No verified release-wide medical pair count. Use an identified study subset and document its source, selection and pairing. | Legacy URL |
| OASIS | Cross-sectional and longitudinal imaging of normal aging and cognitive decline. | OASIS-1: T1 MRI. OASIS-3: T1, T2, FLAIR and other MRI sequences, plus PET; availability varies by participant and visit. | OASIS-1 (2007): 416 participants. OASIS-3 (2019 description): 1,098 participants; this is not a current cumulative total. | OASIS portal<br>2019 OASIS-3 description |
| IXI | Healthy-subject brain MRI for anatomical analysis and registration. | T1, T2, PD, MRA and diffusion MRI. | Approximately 600 healthy-subject MR datasets from three London hospitals; sequence availability must be checked when forming paired inputs. | IXI dataset |
| CHAOS | Healthy abdominal-organ segmentation; CT and MRI cohorts are not paired across modalities. | CT; MRI T1-DUAL (in-/out-of-phase) and T2-SPIR. | 40 CT patients and a separate 40 MRI patients. MRI comprises 120 series (three per patient). CT and MRI are from different, unregistered patients. | Challenge data |
| SynthRAD 2023 | Synthetic CT generation for radiotherapy in brain and pelvis. | Task 1: paired MRI-CT. Task 2: paired CBCT-CT. | 540 pairs per task: 360 training, 60 validation and 120 test pairs. The two tasks total 1,080 pairs; these counts are not a single MRI-CT cohort. | 2023 challenge<br>Challenge design |
| HECKTOR 2021 | Head-and-neck primary-tumor segmentation and outcome prediction. | Registered FDG-PET and CT. | 2021: 224 training and 101 test patient cases (325 total). Each case contains paired PET/CT; 224 is not the full challenge total. | 2021 challenge |

Table 10: Comprehensive evaluation system for MMIF

| Metric or measure | Description |
|---|---|
| **Clinical Qualitative Validation** — Determine clinical acceptability<br>**Professional Physician Assessment**. Assesses clinical acceptability in the tested reader group; judgments depend on expertise and protocol. Preference alone does not establish diagnostic benefit. | |
| Structural detail preservation | Assesses the preservation of anatomical morphology and boundaries for the defined reading task. |
| artifact control | Assesses the suppression of generative hallucinations, edge blurring, and false color injections. |
| clinical auxiliary value | Assesses perceived usefulness for lesion localization or target delineation; diagnostic benefit requires task endpoints. |
| **User Study**. Characterizes the tested users and endpoints. Non-expert ratings support usability or preference; time and confidence alone do not establish diagnostic benefit. | |
| Information completeness | Assesses perceived preservation of source information; suspected omissions require source-image checks. |
| visual comfort | Evaluates whether the contrast and color mapping adhere to natural HVS perception. |
| Reading time and decision confidence | Measures reading time and decision confidence in a specified reader group against a defined comparator, alongside accuracy and error rates. |
| **Objective Quantitative Assessment** — Measure fusion image quality (direct assessment)<br>**Source-referenced**. Evaluates agreement with the specified source modalities, which serve as references rather than as ground-truth fused images; Limitation: Often assumes modality symmetry, which dilutes asymmetric features like sparse PET hotspots. | |
| $\mathrm{SSIM}(x,y)=\frac{2\mu_x\mu_y+C_1}{\mu_x^2+\mu_y^2+C_1}\cdot\frac{2\sigma_{xy}+C_2}{\sigma_x^2+\sigma_y^2+C_2}$ | Compares luminance, contrast and structure. $C_1$ and $C_2$ are positive stabilizing constants. |
| $\mathrm{PSNR}=20\cdot\log_{10}\left(\frac{\mathrm{MAX}_I}{\sqrt{MSE}}\right)$<br>$\mathrm{MSE}=\frac{1}{mn}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}[I(i,j)-K(i,j)]^2$ | Quantifies absolute pixel-level error. Limited in multi-modality fusion due to inherent intensity discrepancies between asymmetric source modalities. |
| $\mathrm{MI}=MI_{A,F}+MI_{B,F}$ | Quantifies the Kullback-Leibler divergence between the joint and marginal distributions, reflecting non-linear information dependence. |
| $\mathrm{CC}=\frac{r(\mathbf{A},\mathbf{F})+r(\mathbf{B},\mathbf{F})}{2}$ | Measures the linear global correlation of pixel intensities between the fused image and the source modalities. |
| $Q^{AB/F}=\frac{\sum_{i=1}^{M}\sum_{j=1}^{N}(Q^{A,F}(i,j)w^A(i,j)+Q^{B,F}(i,j)w^B(i,j))}{\sum_{i=1}^{M}\sum_{j=1}^{N}(w^A(i,j)+w^B(i,j))}$ | Explicitly evaluates the preservation of gradient magnitude and orientation (edge vectors), critical for delineating anatomical boundaries. |
| $\mathrm{N}^{\mathrm{AB/F}}=\frac{\sum_{i=1}^{M}\sum_{j=1}^{N}AM(i,j)[(1-Q^{A,F}(i,j))w^A(i,j)+(1-Q^{B,F}(i,j))w^B(i,j)]}{\sum_{i=1}^{M}\sum_{j=1}^{N}(w^A(i,j)+w^B(i,j))}$ | Quantifies the localized artifacts or noise injected during the fusion process by assessing the normalized absolute bias factor. |
| $\mathrm{RMSE}=\sqrt{\frac{1}{XY}\sum_{i=0}^{X-1}\sum_{j=0}^{Y-1}(R(i,j)-F(i,j))^2}$ | Measures global pixel deviation against an ideal reference or ground truth (if available in synthetic scenarios). |
| $\mathrm{SCD}=r(\mathbf{F}-\mathbf{B},\mathbf{A})+r(\mathbf{F}-\mathbf{A},\mathbf{B})$ | Computes the cross-correlations between each source and the complementary difference image. A higher value indicates stronger transfer of complementary source information, but does not directly detect artifacts or generative hallucinations. |
| **Objective Quantitative Assessment** — Measure fusion image quality (direct assessment)<br>**No-reference (output statistics)**. Evaluates intrinsic richness without reference images; Limitation: Highly susceptible to "metric deception" (can be artificially inflated by noise and artifacts). | |
| $\mathrm{EN}=-\sum_{l=0}^{L-1}p_l\log_2 p_l$ | Calculates the Shannon entropy of the image histogram. Reflects global information capacity and dynamic range, but is completely blind to spatial arrangement. |
| $\mathrm{SF}=\sqrt{RF^2+CF^2}$ | Calculates row and column spatial gradients. Indicates local detail richness and edge sharpness, but cannot distinguish between genuine textures and high-frequency noise. |
| $\mathrm{SD}=\sqrt{\frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(F(i,j)-\mu_F)^2}$ | Computes the dispersion of fused-image intensities around their mean, serving as a first-order statistical proxy for overall image contrast. |
| $\mathrm{AG}=\frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}\sqrt{\frac{\nabla F_x^2(i,j)+\nabla F_y^2(i,j)}{2}}$ | Measures the average magnitude of local pixel differences, reflecting micro-texture sharpness across the entire spatial domain. |
| **Objective Quantitative Assessment** — Measure specified task performance (indirect assessment)<br>**Downstream Tasks**. Measures performance under the tested task model, data, and reference standard. Algorithmic gains alone do not establish clinical benefit. | |
| $\mathrm{Dice}=\frac{2\times TP}{FP+2\times TP+FN}$ | Quantifies the volumetric overlap between predicted and ground-truth regions; optimal for evaluating complete lesion or organ coverage. |

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Table 10: Comprehensive evaluation system for MMIF (continued)

| Metric or measure | Description |
|---|---|
| IoU = $\frac{TP}{FP+TP+FN}$ | Calculates the intersection over union; serves as the fundamental penalty metric for spatial mismatch in detection and segmentation tasks. |
| HD95$(A, B) = \max(d_{95}(A, B), d_{95}(B, A))$ | Calculates the 95th percentile of the shortest distances between two boundary point sets. Unlike maximum HD, it evaluates boundary matching precision while remaining robust to small outlier noise. |
| ACC = $\frac{TP+TN}{TP+TN+FP+FN}$ | Represents the proportion of correct global predictions. While intuitive, it is highly sensitive to class imbalance in clinical datasets. |
| AUC = Area under the ROC curve | Evaluates the classifier's ability to rank positive instances higher than negative ones; highly robust to imbalanced clinical data distributions. |
| mAP = $\frac{1}{N}\sum_{i=1}^{N} AP_i$ | Integrates precision and recall across all classes, comprehensively reflecting detection accuracy and false-positive suppression capabilities. |
| **Engineering-Oriented Evaluation** — Determine clinical deployment feasibility<br>**Efficiency**. Ensures real-world usability and real-time inference; Limitation: Strictly hardware-dependent. | |
| Model size = Total parameters × Number of bytes per parameter | Directly determines the memory footprint (VRAM/RAM) required for loading the fusion model onto clinical edge devices. |
| FLOPs = The sum of the number of floating point operations for all layers in the model | Quantifies the theoretical computational complexity required by the model to complete a single forward inference pass. |
| FPS = $\frac{n}{\text{EndTime}-\text{StartTime}}$ | Measures the empirical throughput (frames processed per second) in practical deployment environments. |
| **Robustness**. Guarantees adaptability to imperfect real-world clinical data; Limitation: Difficult to benchmark all clinical degradation scenarios. | |
| Modality missing tolerance | Evaluates the model's resilience to maintain stable diagnostic feature extraction when one or more input modalities are absent. |
| Noise resistance | Assesses the model's capability to suppress sensor noise, including speckle, Poisson, without destroying underlying anatomical textures. |
| Registration error resilience | Measures the stability of fusion performance under imperfect spatial alignment, a critical necessity for non-rigid anatomical structures. |